



**IEEE Standard for
Local and metropolitan area networks—**

Virtual Bridged Local Area Networks

**Amendment 12: Forwarding and
Queuing Enhancements for
Time-Sensitive Streams**

IEEE Computer Society

Sponsored by the
LAN/MAN Standards Committee

IEEE
3 Park Avenue
New York, NY 10016-5997, USA

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IEEE Std 802.1Qav™-2009
(Amendment to
IEEE Std 802.1Q™-2005)

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Approved 9 December 2009

IEEE-SA Standards Board

Abstract: This amendment to IEEE Std 802.1Q defines enhancements to the forwarding and queueing functions of a VLAN Bridge to support the transmission of time-sensitive data streams.

Keywords: Bridged Local Area Networks, local area networks (LANs), MAC Bridges, metropolitan area networks, time-sensitive data streams, Virtual Bridged Local Area Networks (virtual LANs)

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This introduction is not part of IEEE Std 802.1Qav-2009, IEEE Standards for Local and Metropolitan Area Networks—Virtual Bridged Local Area Networks—Amendment 12: Forwarding and Queuing Enhancements for Time-Sensitive Streams.

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IEEE Standard for

Local and metropolitan area networks—

Virtual Bridged Local Area Networks

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Editorial Note

This amendment specifies changes to the forwarding and queueing functions described in IEEE Std 802.1Q™. Changes are applied to the base text of IEEE Std 802.1Q-2005, as modified by those amendments that had been approved, but not incorporated into the base text of the standard, at the time that this amendment was approved, namely (in chronological order) IEEE Std 802.1ad™, IEEE Std 802.1ak™, IEEE Std 802.1ag™, IEEE Std 802.1ah™, IEEE Std 802.1Q-2005/Cor 1, IEEE Std 802.1ap™, IEEE Std 802.1Qaw™, IEEE Std 802.1Qay™, and IEEE Std 802.1aj™. Text shown in bold italics in this amendment defines the editing instructions necessary to changes to this base text. Three editing instructions are used: **change**, **delete**, and **insert**. **Change** is used to make a change to existing material. The editing instruction specifies the location of the change and describes what is being changed. Changes to existing text may be clarified using ~~strikeout~~ markings to indicate removal of old material, and underline markings to indicate addition of new material. **Delete** removes existing material. **Insert** adds new material without changing the existing material. Insertions may require renumbering. If so, renumbering instructions are given in the editing instruction. Editorial notes will not be carried over into future editions of IEEE Std. 802.1Q.

1. Overview

1.1 Scope

Insert the following text at the end of 1.1, renumbering the list items and NOTES as necessary:

This standard allows bridges to provide performance guarantees for time-sensitive (i.e., bounded latency and latency variation) loss-sensitive real-time audio video (AV) data stream transmission (AV traffic). It specifies priority regeneration and controlled bandwidth queue draining algorithms. Virtual Local Area Network (VLAN) tag encoded priority values are allocated, in aggregate, to segregate frames among queues that support AV traffic and queues that support non-AV traffic, allowing simultaneous support of both AV traffic and other bridged traffic over and between wired and wireless Local Area Networks (LANs). To this end, it

- a) Defines status parameters that allow the boundaries of a Stream Reservation Protocol (SRP—see Clause 35¹) domain (6.6.4) to be identified and maintained
- b) Specifies how the priority information in frames received at SRP domain boundary ports is regenerated

NOTE 1—The priorities in frames transmitted from outside an SRP domain to a Bridge inside an SRP domain are re-mapped in order to ensure that traffic that is not associated with a reservation does not disrupt traffic that is associated with a reservation. Hence, traffic entering an SRP domain that uses priority code point values associated with reserved traffic classes will be re-mapped to priority code point values that are not associated with reserved traffic classes.²

- c) Specifies how priority information is used to determine the traffic classes to be used for time-sensitive streams
- d) Defines a credit-based shaper algorithm to shape traffic in accordance with stream reservations

NOTE 2—The credit-based shaper algorithm operates on the outbound queues; the mechanisms specified for the support of time-sensitive AV traffic do not involve any form of ingress metering or policing.

¹Clause 35 is defined in IEEE P802.1Qat™ [B31] that was a draft standard under development at the time of publication.

²Notes in text, tables, and figures are given for information only and do not contain requirements needed to implement the standard.

2. Normative references

Insert the following references into Clause 2 in appropriate collating sequence:

IEEE Std 802.1AX™-2008, IEEE Standard for Local and Metropolitan Area Networks—Link Aggregation.

3. Definitions

Insert the following text immediately after the first paragraph of Clause 3:

This standard makes use of the following terms defined in IEEE Std 802:

- end station
- station

Insert the following definitions into Clause 3, in alphabetical order, renumbering existing definitions as needed:

3.1 audio video (AV) traffic: Traffic associated with audio and/or video applications that is sensitive to transmission latency and latency variation, and, for some applications, packet loss.

3.2 latency: The delay experienced by a frame in the course of its propagation between two points in a network, measured from the time that a known reference point in the frame passes the first point to the time that the reference point in the frame passes the second point.

NOTE 1—Latency is sometimes referred to as *frame delay*.

NOTE 2—The term *frame* is used in this standard to refer to a data frame, and not to a video frame.

3.3 stream reservation (SR) class: A traffic class whose bandwidth can be reserved for AV traffic. A priority value is associated with each SR class. SR classes are denoted by consecutive letters of the alphabet, starting with A and continuing for up to seven classes.

3.4 Stream Reservation Protocol (SRP) domain boundary port: A Port of a station that is a member of an SRP domain for a given SR class, and that is connected via an individual LAN to a Port of a station that is either within a different SRP domain for that SR class, or is not part of any SRP domain.

NOTE—The term *SRP domain* is defined in 6.6.4.

3.5 Stream Reservation Protocol (SRP) domain core port: A Port of a station that is a member of an SRP domain for a given SR class, and that is connected, via an individual LAN that is part of the active topology, to a Port of a station that is a member of the same SRP domain for that SR class.

NOTE—The term *SRP domain* is defined in 6.6.4.

3.6 time-sensitive stream: A stream of data frames that are required to be delivered with a bounded latency.

4. Abbreviations

Insert the following abbreviations into Clause 4 in alphabetical order:

AV	audio/video
AVB	audio/video bridging
CSN	Coordinated Shared Network
FQTSS	Forwarding and Queuing Enhancements for time-sensitive streams
SR	stream reservation
SRP	Stream Reservation Protocol
TSpec	traffic specification

5. Conformance

5.4.1 VLAN-aware bridge component options

Insert new bullet item p) as shown, following existing bullet item o):

- p) Support forwarding and queuing for time-sensitive streams (5.4.1.5).

Insert new 5.4.1.5 as shown, following existing 5.4.1.4:

5.4.1.5 Forwarding and queuing for time-sensitive streams—requirements

A VLAN-aware Bridge component implementation that conforms to the provisions of this standard for forwarding and queuing for time-sensitive streams shall:

- a) Support a minimum of two traffic classes on all Ports, of which
 - 1) A minimum of one traffic class supports the strict priority algorithm for transmission selection (8.6.8.1), and
 - 2) One traffic class is an SR class.
- b) Support the operation of the credit-based shaper algorithm (8.6.8.2) on all Ports as the transmission selection algorithm used for the SR class.
- c) Support SRP domain boundary port priority regeneration override as defined in 6.9.4, and the default priority regeneration override value defined in Table 6-6, for SR class “B”.
- d) Support the tables and procedures for mapping priorities to traffic classes as defined in 34.5.

A VLAN-aware Bridge component implementation that conforms to the provisions of this standard for forwarding and queuing for time-sensitive streams may:

- e) Support the management entities defined in 12.21 by means of the SNMP MIB module defined in 17.7.13.
- f) Support two or more SR classes (a maximum of seven), and support the operation of the credit-based shaper algorithm (8.6.8.2) on all Ports as the transmission selection algorithm used for those SR classes. The number of SR classes supported shall be stated in the PICS.
- g) Support SRP domain boundary port priority regeneration override as defined in 6.9.4, and the default priority regeneration override value defined in Table 6-6, for SR class “A”. If more than two SR classes are supported, the default priority regeneration override values used for the additional SR classes shall be stated in the PICS.

Insert the following new subclause after existing 5.17, renumbering as necessary:

5.18 End station requirements—forwarding and queuing for time-sensitive streams

An end station implementation that conforms to the provisions of this standard for forwarding and queuing for time-sensitive streams shall:

- a) Support a minimum of two traffic classes on all Ports, of which
 - 1) A minimum of one traffic class supports the strict priority algorithm for transmission selection (8.6.8.1), and
 - 2) One traffic class is an SR class.
- b) Support the operation of the credit-based shaper algorithm (8.6.8.2) as the transmission selection algorithm used for frames transmitted for each stream associated with the SR class.
- c) Support the operation of the credit-based shaper algorithm (8.6.8.2) on all Ports as the transmission selection algorithm used for the SR class.

- d) Use the default priority associated with SR class “B” as shown in Table 6-6 as the priority value carried in transmitted SR class “B” data frames.

An end station implementation that conforms to the provisions of this standard for forwarding and queuing for time-sensitive streams may:

- e) Support two or more SR classes (a maximum of seven), and support the operation of the credit-based shaper algorithm (8.6.8.2) on all Ports as the transmission selection algorithm used for those SR classes. The number of SR classes supported shall be stated in the PICS.
- f) Use the default priority associated with SR class “A” as shown in Table 6-6 as the priority value carried in transmitted SR class “A” data frames. If more than two SR classes are supported, the priority value carried in transmitted data frames for the additional SR classes shall be stated in the PICS.

6. Support of the MAC service

6.6 Internal Sublayer Service

Insert new 6.6.4 as follows:

6.6.4 Stream Reservation Protocol (SRP) Domain status parameters

A Stream Reservation Protocol (SRP) domain is a set of stations (end stations and/or Bridges), their Ports, and the attached individual LANs, that satisfy all of the following conditions for a given SR class:

- a) Those stations that transmit streams all support the credit-based shaper algorithm, defined in 8.6.8, as the transmission selection method for the SR class.
- b) The stations all support SRP, as defined in Clause 35, as the means of creating bandwidth reservations for the SR class.
- c) Those stations that transmit streams all associate the same priority value with the SR class.
- d) Each Port in the set is either an SRP domain core port or an SRP domain boundary port.
- e) Each SRP domain core Port in the set is connected, via an individual LAN that is part of the active topology, to an SRP domain core Port of another station in the set.

In Bridges that support the Stream Reservation Protocol (SRP), and for each SR class supported by the Bridge, an **SRPdomainBoundaryPort** parameter is associated with each Port of the Bridge.

Each SRPdomainBoundaryPort parameter has a Boolean value. The value of the SRPdomainBoundaryPort parameter for a given SR class is TRUE if the operation of SRP has determined that the Port is an SRP domain boundary port (3.4) for that SR class; otherwise, the Port is either an SRP domain core port (3.5) for that SR class or is not part of that SRP domain, and the SRPdomainBoundaryPort parameter value is FALSE.

6.9 Support of the EISS

6.9.4 Regenerating priority

Change 6.9.4, and insert new Table 6-6, renumbering subsequent tables, as follows:

The priority of each received frames is regenerated using priority information contained in the frame and the Priority Regeneration Table for the reception Port. For each reception Port, the Priority Regeneration Table has eight entries, corresponding to the eight possible values of priority (0 through 7). Each entry specifies, for the given value of received priority, the corresponding regenerated value.

For untagged frames, the priority parameter signaled in the corresponding M_UNITDATA.indication is taken to be the received priority. For tagged frames, the priority signaled in the PCP field of the tag header is taken to be the received priority.

NOTE 1—IEEE 802 LAN technologies signal a maximum of eight priority values. Annex G further explains the use of priority values and how they map to traffic classes.

Table 6-5 specifies default regenerated priority values for each of the eight possible values of the received priority. These default values shall be used as the initial values of the corresponding entries of the Priority Regeneration Table for each Port.

The values in the Priority Regeneration Table may be modified by management, as described in Clause 12. If this capability is provided, the value of the table entries shall be independently settable for each reception

Port and for each value of received priority, and the Bridge shall have the capability to use the full range of values in the parameter ranges specified in Table 6-5.

NOTE 2—The regeneration and mapping of priority within the Bridge should be consistent with the end-to-end significance of that priority across the rest of the Bridged Local Area Network. The regenerated priority value is used:

- Via the traffic class table (8.6.6) to determine the traffic class for a given outbound Port, and
- Via fixed, MAC type-specific mappings (6.7) to determine the access priority that will be used for certain media access methods.

Table 6-5—Priority regeneration

Received priority	Default regenerated priority	Range
0	0	0–7
1	1	0–7
2	2	0–7
3	3	0–7
4	4	0–7
5	5	0–7
6	6	0–7
7	7	0–7

In Bridges that support SRP, an SRP domain boundary port priority regeneration override table is maintained for each reception Port. This table associates a priority value with each SR class supported by the Bridge, and specifies the priority value to be used as the regenerated priority, in preference to the value in the Priority Regeneration table, when the SRPdomainBoundaryPort parameter for that SR class has the value TRUE. Table 6-6 specifies the default values for priority and regenerated priority for two SR classes, SR class A and SR class B. In cases where only SR class B is supported, only the default value shown in the table for SR Class B is used.

Table 6-6—Default SRP domain boundary port priority regeneration override values

<u>SR class</u>	<u>Default priority</u>	<u>Default regenerated priority for SRP domain boundary Ports</u>	<u>Range</u>
A	<u>3</u>	<u>0</u>	<u>0–7</u>
B	<u>2</u>	<u>0</u>	<u>0–7</u>

The priority values contained in the SRP domain boundary port priority regeneration override table may be modified by management, as described in Clause 12. If this capability is provided, the value of the table entries shall be independently settable for each reception Port and for each supported SR class, and the Bridge shall have the capability to use the full range of values in the parameter ranges specified in Table 6-6.

NOTE 3—The default values specified for the SRP domain boundary port Priority Regeneration Table map inbound PCP values onto the next lowest priority that is available, leaving the remaining priority values unchanged. The consequence of this remapping is that traffic entering an SRP domain that carries the priority value associated with that domain is forwarded by the bridge at the domain boundary using a lower priority. There is no means of mapping these priority values back to their original values as frames exit from an SRP domain. The default mappings are based on the assumption that normally, two SR classes (A and B) are used to support traffic for which bandwidth has been reserved; if only the lower class (B) is supported, then the default override value for the unsupported A class does not apply.

NOTE 4—The ability for the priority regeneration table to be freely modified by management means that it is possible to create mappings that result in reserved and non-reserved traffic sharing the same priority. This will have unpredictable effects upon the behavior of the network with respect to the worst-case latency that can be assumed for reserved traffic.

8. Principles of bridge operation

8.6 The Forwarding Process

8.6.6 Queuing frames

Change the text of 8.6.6 as follows:

The Forwarding Process provides storage for queued frames, awaiting an opportunity to submit these for transmission. The order of frames received on the same Bridge Port shall be preserved for

- a) Unicast frames with a given VID, priority, and destination address and source address combination;
- b) Multicast frames with a given VID, priority, and destination address.

The Forwarding Process provides one or more queues for a given Bridge Port, each corresponding to a distinct traffic class. Each frame is mapped to a traffic class using the Traffic Class Table for the Port and the frame's priority. Traffic class tables may be managed. Table 8-3 shows the recommended mapping for the number of classes implemented, in implementations that do not support the credit-based shaper transmission selection algorithm (8.6.8.2). The requirements for priority to traffic class mappings in implementations that support the credit-based shaper transmission selection algorithm are defined in 34.5. Up to eight traffic classes may be supported, allowing separate queues for each priority.

Table 8-3—Recommended priority to traffic class mappings

		Number of Available Traffic Classes							
		1	2	3	4	5	6	7	8
Priority	0 (Default)	0	0	0	0	0	1	1	1
	1	0	0	0	0	0	0	0	0
	2	0	0	0	1	1	2	2	2
	3	0	0	0	1	1	2	3	3
	4	0	1	1	2	2	3	4	4
	5	0	1	1	2	2	3	4	5
	6	0	1	2	3	3	4	5	6
	7	0	1	2	3	4	5	6	7

NOTE—The rationale for these mappings is discussed in Annex G (informative).

NOTE—Different numbers of traffic classes may be implemented for different Ports. Ports with media access methods that support a single transmission priority, such as CSMA/CD, can support more than one traffic class.

Delete existing 8.6.8 and insert new 8.6.8, 8.6.8.1, 8.6.8.2, and Table 8-4, renumbering subsequent tables as necessary, as follows:

8.6.8 Transmission selection

For each Port, frames are selected for transmission on the basis of the traffic classes that the Port supports and the operation of the transmission selection algorithms supported by the corresponding queues on that Port. For a given Port and supported value of traffic class, frames are selected from the corresponding queue for transmission if and only if

- a) The operation of the transmission selection algorithm supported by that queue determines that there is a frame available for transmission; and
- b) For each queue corresponding to a numerically higher value of traffic class supported by the Port, the operation of the transmission selection algorithm supported by that queue determines that there is no frame available for transmission.

The strict priority transmission selection algorithm defined in 8.6.8.1 shall be supported by all Bridges as the default algorithm for selecting frames for transmission. The credit-based shaper transmission selection algorithm defined in 8.6.8.2 may be supported in addition to the strict priority algorithm. Further transmission selection algorithms, selectable by management means, may be supported as an implementation option so long as the requirements of 8.6.6 are met.

The Transmission Selection Algorithm Table for a given Port assigns, for each traffic class that the Port supports, the transmission selection algorithm that is to be used to select frames for transmission from the corresponding queue. Transmission Selection Algorithm Tables may be managed, and allow the identification of vendor-specific transmission selection algorithms. The transmission selection algorithms are identified in the Transmission Selection Algorithm Table by means of integer identifiers, as defined in Table 8-4.

Table 8-4—Transmission selection algorithm identifiers

Transmission selection algorithm	Identifier
Strict priority (8.6.8.1)	0
Credit-based shaper (8.6.8.2)	1
Reserved for future standardization	2–255
Vendor-specific	A four-octet integer, where the 3 most significant octets hold an OUI value, and the least significant octet holds an integer value in the range 0–255 assigned by the owner of the OUI.

In implementations that do not support the credit-based shaper transmission selection algorithm, the recommended default configuration for the Transmission Selection Algorithm Tables is to assign the strict priority transmission selection algorithm to all supported traffic classes. In implementations that support the credit-based shaper transmission selection algorithm, the recommended default configuration for the Transmission Selection Algorithm Tables is defined in 34.5.

8.6.8.1 Strict priority algorithm

For a given queue that supports strict priority transmission selection, the algorithm determines that there is a frame available for transmission if the queue contains one or more frames. The order in which frames are selected for transmission from the queue shall maintain the ordering requirement specified in 8.6.6.

8.6.8.2 Credit-based shaper algorithm

For a given queue that supports credit-based shaper transmission selection, the algorithm determines that a frame is available for transmission if the following conditions are all true:

- a) The queue contains one or more frames.
- b) The *transmitAllowed* signal for the queue is TRUE.

The order in which frames are selected for transmission from the queue shall maintain the ordering requirement specified in 8.6.6.

The following external parameters are associated with each queue that supports the operation of the credit-based shaper algorithm:

- c) ***portTransmitRate***. The transmission rate, in bits per second, that the underlying MAC service that supports transmission through the Port provides. The value of this parameter is determined by the operation of the MAC.
- d) ***idleSlope***. The rate of change of *credit*, in bits per second, when the value of *credit* is increasing (i.e., while *transmit* is FALSE). The value of *idleSlope* can never exceed *portTransmitRate*. The value of *idleSlope* for a given queue is determined by the value of the ***operIdleSlope(N)*** parameter for that queue, as defined in 34.3.

The following internal parameters are associated with each queue that supports the credit-based shaper algorithm:

- e) ***transmit***. Takes the value TRUE for the duration of a frame transmission from the queue; FALSE when any frame transmission from the queue has completed.
- f) ***credit***. The transmission credit, in bits, that is currently available to the queue. If, at any time, there are no frames in the queue, and the *transmit* parameter is FALSE, and *credit* is positive, then *credit* is set to zero.
- g) ***sendSlope***. The rate of change of *credit*, in bits per second, when the value of *credit* is decreasing (i.e., while *transmit* is TRUE). The value of *sendSlope* is defined as follows:

$$\text{sendSlope} = (\text{idleSlope} - \text{portTransmitRate})$$

- h) ***transmitAllowed***. Takes the value TRUE when the *credit* parameter is zero or positive; FALSE when the *credit* parameter is negative.

NOTE 1—The value of *credit* is naturally constrained by the operating parameters for the algorithm, and also by the operation of any higher priority queues that use this algorithm. The lowest value that *credit* can reach depends on *sendSlope* and the largest frame that will be transmitted from the queue. This largest frame size is a consequence of the type of traffic that is being transmitted using the queue, rather than a limit imposed by the algorithm or by management. The highest value that can be accumulated in *credit* depends on *idleSlope* and the length of time that the algorithm may have to wait when the queue is not empty and there is other traffic being transmitted through the Port; for any given queue, that length of time is bounded, as discussed in Annex L, which also illustrates how the algorithm operates under varying conditions, and how its operation affects the maximum latency that can be experienced by SR traffic.

NOTE 2—In order for the credit-based shaper algorithm to operate as intended, it is necessary for all traffic classes that support the algorithm to be numerically higher than any traffic classes that support the strict priority algorithm defined in

8.6.8.1. The mapping of priorities onto traffic classes, and the choice of traffic classes that support particular transmission selection algorithms, is defined in 34.5.

8.8 The Filtering Database

Change the text of 8.8 as follows:

The Filtering Database supports frame filtering (8.6.3) queries by the Forwarding Process to determine whether received frames, with given values of destination MAC Address and VID, are to be forwarded through a given potential transmission Port (8.6.2, 8.6.3, 8.6.4). ~~¶ The Filtering Database contains filtering information in the form of filtering entries that are either~~

- a) Static, and explicitly configured by management action; or
- b) Dynamic, and automatically entered into the Filtering Database by the normal operation of the bridge and the protocols it supports.

Two entry types are used to represent static filtering information. The Static Filtering Entry represents static information in the Filtering Database for individual and for group MAC Addresses. It allows administrative control of

- c) Forwarding of frames with particular destination addresses; and
- d) The inclusion in the Filtering Database of dynamic filtering information associated with Extended Filtering Services, and use of this information.

The Filtering Database shall contain entries of the Static Filtering Entry type.

The Static VLAN Registration Entry represents all static information in the Filtering Database for VLANs. It allows administrative control of

- e) Forwarding of frames with particular VIDs;
- f) The inclusion/removal of tag headers in forwarded frames; and
- g) The inclusion in the Filtering Database of dynamic VLAN membership information, and use of this information.

The Filtering Database may contain entries of the Static VLAN Registration Entry type.

Static filtering information is added to, modified, and removed from the Filtering Database only under explicit management control. It shall not be automatically removed by any ageing mechanism. Management of static filtering information may be carried out by use of the remote management capability provided by Bridge Management (8.12) using the operations specified in Clause 12.

~~Three~~Four entry types are used to represent dynamic filtering information:

- h) ~~Each Dynamic Filtering Entry is used to specify the Ports on through which a frame with a given individual source MAC Address and VID has been or can be received. have been learned for a given VLAN. They are Dynamic Filtering Entries can be created and updated by the Learning Process (8.7). Entries that are updated by the Learning Process and are subject to ageing and removal by the Filtering Database.~~
- i) MAC Address Registration Entries support the registration of ~~group~~ MAC Addresses. They are created, updated, and removed by the MMRP protocol in support of Extended Filtering Services (8.8.4, ~~6.6.5~~ and Clause 10 of IEEE Std 802.1D), ~~subject to the state of the Restricted_Group_Registration management control (10.3.2.3 of IEEE Std 802.1D). If the value of this the Restricted_MAC_Address_Registration management control (10.12.2.3) is TRUE, then the~~

creation of a MAC Address Registration Entry is not permitted unless a Static Filtering Entry exists that permits dynamic registration for the MAC address concerned.

- j) Dynamic VLAN Registration Entries are used to specify the Ports on which VLAN membership has been dynamically registered. They are created, updated, and removed by the MVRP protocol, in support of automatic VLAN membership configuration (Clause 11), subject to the state of the Restricted_VLAN_Registration management control (11.2.3.2.3). If the value of this control is TRUE, then the creation of a Dynamic VLAN Registration Entry is not permitted unless a Static VLAN Registration Entry exists that permits dynamic registration for the VLAN concerned.
- k) Dynamic Reservation Entries are used to specify the Ports on which stream reservations have been made. They are created, updated, and removed by the operation of SRP defined in Clause 35.

Static Filtering Entries and MAC Address Registration Entries comprise

- l) A MAC Address specification;
- m) A VLAN Identifier (VID);
- n) A Port Map, with a control element for each outbound Port to specify filtering for that MAC Address specification and VID.

Dynamic Filtering Entries comprise

- o) A MAC Address specification;
- p) A locally significant Filtering Identifier (FID; see 8.8.9);
- q) A Port Map, with a control element for each outbound Port to specify filtering for that MAC Address specification in the VLAN(s) allocated to that FID.

The Port Map in Dynamic Filtering Entries may contain a **connection_identifier** (8.8.11) for each outbound Port.

Static and Dynamic VLAN Registration Entries comprise

- r) A VLAN Identifier;
- s) A Port Map, with a control element for each outbound Port to specify filtering for the VLAN.

Dynamic Reservation Entries comprise

- t) A MAC Address specification;
- u) A VID;
- v) A Port Map, with a control element for each outbound Port to specify filtering for that MAC Address specification in the VLAN specified by the VLAN Identifier.

Dynamic filtering information may be read by use of the remote management capability provided by Bridge Management (8.12) using the operations specified in Clause 12.

The Filtering Services supported by a Bridge (Basic and Extended Filtering Services) determine the default behavior of the Bridge with respect to the forwarding of frames destined for group MAC Addresses. In Bridges that support Extended Filtering Services, the default forwarding behavior for group MAC Addresses, for each Port, and for each VID, can be configured both statically and dynamically by means of Static Filtering Entries and/or MAC Address Registration Entries that can carry the following MAC Address specifications:

- w) All Group Addresses, for which no more specific Static Filtering Entry exists;
- x) All Unregistered Group Addresses (i.e., all group MAC Addresses for which no MAC Address Registration Entry exists), for which no more specific Static Filtering Entry exists.

NOTE 1—The All Group Addresses specification in item ~~sw~~), when used in a Static Filtering Entry with an appropriate control specification, provides the ability to configure a Bridge that supports Extended Filtering Services to behave as a Bridge that supports only Basic Filtering Services on some or all of its Ports. This might be done for the following reasons:

- The Ports concerned serve “legacy” devices that wish to receive multicast traffic, but are unable to register Group membership;
- The Ports concerned serve devices that need to receive all multicast traffic, such as routers or diagnostic devices.

The Filtering Database shall support the creation, updating, and removal of Dynamic Filtering Entries by the Learning Process (8.7). In Bridges that support Extended Filtering Services, the Filtering Database shall support the creation, updating, and removal of MAC Address Registration Entries by MMRP (Clause 10).

In Bridges that support SRP, the Filtering Database shall support the creation, updating, and removal of Dynamic Reservation Entries by SRP (Clause 35).

Figure 8-3 illustrates use of the Filtering Database by the Forwarding Process in a single instance of frame relay between the Ports of a Bridge with two Ports.

Figure 8-4 illustrates the creation or update of a dynamic entry in the Filtering Database by the Learning Process. The entries in the Filtering Database allow MAC Address information to be learned independently for each VLAN or set of VLANs, by relating a MAC Address to the VLAN or set of VLANs on which that address was learned. ~~This has the effect of creating independent Filtering Databases for each VLAN or set of VLANs that is present in the network.~~

~~NOTE 2—This standard specifies a single Filtering Database that contains all Filtering Database entries; however, the inclusion of VIDs and FIDs in the filtering entries effectively provides distinct IEEE Std 802.1D-style Filtering Databases per VLAN or set of VLANs.~~

NOTE 3~~2~~—The ability to create VLAN-dependent Filtering Database entries allows a Bridge to support

- Multiple end stations with the same individual MAC Address residing on different VLANs;
- End stations with multiple interfaces, each using the same individual MAC Address, as long as not more than one end station or interface that uses a given MAC Address resides in a given VLAN.

Figure 8-5 illustrates the operation of the Spanning Tree Protocol Entity (8.10), which operates the Spanning Tree Algorithm and Protocol, and its notification of the Filtering Database of changes in active topology signaled by that protocol.

There are no standardized constraints on the size of the Filtering Database in an implementation for which conformance to this standard is claimed. The PICS Proforma in Annex A requires the following to be specified for a given implementation:

- y) The total number of entries (Static Filtering Entries, Dynamic Filtering Entries, MAC Address Registration Entries, Static VLAN Registration Entries, ~~and~~ Dynamic VLAN Registration Entries, and Dynamic Reservation Entries) that the implementation of the Filtering Database can support, and
- z) Of that total number, the total number of VLAN Registration Entries (Static and Dynamic) that the Filtering Database can support.

Insert new 8.8.7, as shown, renumbering as necessary:

8.8.7 Dynamic Reservation Entries

A Dynamic Reservation Entry specifies the following:

- a) A group MAC address or an individual MAC address.
- b) The VID of the VLAN in which the dynamic reservation information was registered.
- c) A Reservation Port Map, consisting of a control element for each outbound Port that specifies forwarding (reservation information exists for this Port) or filtering (reservation information does not exist for this Port) of frames for this Port. The initial value shall be filtering.

Dynamic Reservation Entries are created, modified, and deleted by the operation of SRP. No more than one Dynamic Reservation Entry shall be created in the Filtering Database for a given combination of MAC Address and VID.

8.8.9 ~~8.8.8~~ Querying the Filtering Database

Change the text of 8.8.9, and insert new Table 8-8, renumbering subsequent tables, as follows:

If the VID assigned to a relayed frame identifies a VLAN or an ESP with a given outbound Bridge Port in its member set (8.8.9), the Filtering Database entries that are applicable to the frame's destination MAC Address and VID determine whether the frame is filtered or forwarded through that Bridge Port. More than one entry can apply to a given frame. This clause (~~8.8.8~~ 8.8.9) specifies the effect of combining applicable entries, and of the absence of certain types of entries (not all possible entries are necessarily present). For an overview of the different types of entries see the introductory text of ~~clause~~ 8.8.

The Filtering Database entries applicable to a frame whose destination MAC Address is a specific Individual MAC Address are: the Dynamic Filtering Entry (if present) with that MAC Address and the FID to which the frame's VID is allocated (8.8.7), and all Static Filtering Entries (if any) with that MAC Address, or with "All Individual MAC Address", as their MAC Address specification and a VID that is also allocated to that FID, as their VLAN identifier specification. An entry with a wildcard VID applies only if there is no applicable Static Filtering Entry for a specific VID. Table 8-5 specifies the result of combining this information for an individual MAC address and a given outbound Bridge Port, and can be summarized as follows:

- IF any static entry for a VID allocated to the FID specifies Forward THEN Forward;
- ELSE IF any static entry for a VID allocated to the FID specifies Filter THEN Filter;
- ELSE IF a static entry for the wildcard VID specifies Forward THEN Forward;
- ELSE IF a static entry for the wildcard VID specifies Filter THEN Filter;
- ELSE IF dynamic (learnt filtering information) entry for the FID specifies Filter THEN Filter;
- ELSE Forward.

Table 8-5—Combining Static and Dynamic Filtering Entries for an individual MAC Address

Filtering Information	Control Elements in any Static Filtering Entry or Entries for this individual MAC Address, FID, and outbound Port specify:				
	Forward (Any Static Filtering Entry for this Address/FID/Port specifies Forward)	Filter (No Static Filtering Entry for this Address/FID/Port specifies Forward)	Use Dynamic Filtering Information (No Static Filtering Entry for this Address/FID/Port specifies Forward or Filter), or no Static Filtering Entry present. Dynamic Filtering Entry Control Element for this individual MAC Address, FID and outbound Port specifies:		
			Forward	Filter	No Dynamic Filtering Entry present
Result	Forward	Filter	Forward	Filter	Forward

The Filtering Database entries applicable (if present) to a frame whose destination MAC Address is a specific group MAC Address are the Static Filtering Entries and MAC Address Registration Entries (for the frame's VID and for the wildcard VID) whose address specification is that group MAC address or "All Group Addresses" or "All Unregistered Group Addresses." A Static Filtering Entry for a wildcard VID applies only if there is no applicable Static Filtering Entry for a specific VID, MAC Address Registration Entries do not use wildcard VIDs. Table 8-6 specifies the results, Registered or Not Registered for a given outbound Bridge Port, of combining the entries for the "All Group Addresses" and for combining the entries for "All Unregistered Group Addresses." Table 8-7 combines these results with the entries for the specific group MAC Address. The result of Table 8-6 for "All Group Addresses" can be summarized as follows:

- IF a static entry for "All Group Addresses" and the frame's VID specifies Forward (Registration Fixed) THEN "All Group Addresses" is Registered;
- ELSE IF a static entry for "All Group Addresses" and the frame's VID specifies Filter (Registration Forbidden) THEN "All Group Addresses" is Not Registered;
- ELSE IF a dynamic (MAC Address Registration) entry for "All Group Addresses" and the frame's VID specifies Forward (Registered) THEN "All Group Addresses" is Registered;
- ELSE "All Group Addresses" is NOT Registered.

NOTE 1—The result for "All Unregistered Group Addresses" is given by substituting that address specification for "All Group Address" throughout the summary.

Table 8-6—Combining Static Filtering Entry and MAC Address Registration Entry for "All Group Addresses" and "All Unregistered Group Addresses"

Filtering Information	Static Filtering Entry Control Element for this group MAC Address, VID, and outbound Port specifies:				
	Registration Fixed (Forward)	Registration Forbidden (Filter)	Use MAC Address Registration Information, or no Static Filtering Entry present. MAC Address Registration Entry Control Element for this group MAC Address, VID and outbound Port specifies:		
			Registered (Forward)	Not Registered (Filter)	No Group Registration Entry present
Result	Registered	Not Registered	Registered	Not Registered	Not Registered

The result of Table 8-7 can be summarized as follows:

- IF a static entry for the specific group address and the frame's VID specifies Forward THEN Forward;
- ELSE IF a static entry for the specific group address and the frame's VID specifies Filter THEN Filter;
- ELSE IF a static entry for the specific group address and the wildcard VID specifies Forward THEN Forward;
- ELSE IF a static entry for the specific group address and the wildcard VID specifies Filter THEN Filter;
- ELSE IF the Table 8-6 result for "All Group Addresses" is Registered THEN Forward;
- ELSE IF the Table 8-6 result for "All Unregistered Group Addresses" is Registered THEN Forward;
- ELSE IF a dynamic (MAC Address Registration) entry for the specific group address and the frame's VID specifies Forward THEN Forward;
- ELSE Filter.

Table 8-7—Forwarding or Filtering for specific group MAC Addresses

				Static Filtering Entry Control Element for this group MAC Address, VID and outbound Port specifies:				
				Registration Fixed (Forward)	Registration Forbidden (Filter)	Use Group Registration Information, or no Static Filtering Entry present. Group Registration Entry Control Element for this group MAC Address, VID and outbound Port specifies:		
						Registered (Forward)	Not Registered (Filter)	No Group Registration Entry present
All Group Addresses control elements for this VID and Port specify (Table 8-6):	Not Registered	All Unregistered Group Addresses control elements for this VID and Port specify (Table 8-6):	Not Registered	Forward	Filter	Forward	Filter (Filter Unregistered Groups)	Filter (Filter Unregistered Groups)
			Registered	Forward	Filter	Forward	Forward (Forward Unregistered Groups)	Forward (Forward Unregistered Groups)
	Registered		Forward	Filter	Forward (Forward All Groups)	Forward (Forward All Groups)	Forward (Forward All Groups)	

When forwarding or filtering a frame with a destination group MAC Address, a VLAN-aware Bridge may:

- Ignore the allocation of VIDs to FID, and use Table 8-7 directly for the frame's VID; or
- Take the same decision for all VIDs allocated to any given FID, forwarding if Table 8-7 specifies Forward for any VID allocated to the same FID as the frame's VID, and filtering otherwise.

For a given destination MAC address and VID, a Dynamic Reservation Entry (8.8.7) may be present in the filtering database, indicating, by means of its port map, which Ports have valid bandwidth reservations. If a Dynamic Reservation Entry exists in the filtering database for that MAC address and VID, the port map in the Dynamic Reservation Entry is used to prevent frames being forwarded on Ports where no reservation exists. Table 8-8 combines the output of Table 8-5 (for an individual MAC address) or Table 8-7 (for a group MAC address) with a Dynamic Reservation Entry to specify forwarding, or filtering, of a frame with that destination MAC Address and VID through an outbound Port.

NOTE 2—Where a group MAC address is used as the destination address for reserved traffic, the reservation mechanisms used by SRP will prevent reservations being made for multiple streams using the same combination of MAC address and VID. The Dynamic Reservation Entry can then control which Port(s) the stream associated with the MAC address and VID is permitted to be transmitted through. The restriction preventing multiple streams from using the same group MAC address/VID is necessary because those different streams may be destined for different destination Ports; if the restriction was not imposed, it would not be possible to distinguish which frames belong to which streams, and therefore, through which Ports they should be transmitted.

Table 8-8—Forwarding or Filtering with Dynamic Reservation Entries

		<u>Dynamic Reservation Entry Control Element for this individual or group MAC Address. VID and outbound Port specifies:</u>	
		<u>Forward</u>	<u>Filter</u>
<u>Output of Table 8-5 (individual MAC address) or Table 8-7 (group MAC address) for this group or individual MAC Address, VID and outbound Port specifies:</u>	<u>Forward</u>	<u>Forward</u>	<u>Filter</u>
	<u>Filter</u>	<u>Filter</u>	<u>Filter</u>

12. Bridge management

Insert the following text and table as new 12.21, immediately following 12.20, renumbering as necessary:

12.21 Management entities for forwarding and queueing for time-sensitive streams

The Bridge enhancements for support of forwarding and queueing for time-sensitive streams are defined in Clause 34.

The objects that comprise this managed resource are as follows:

- a) The Bandwidth Availability Parameter Table (12.21.1)
- b) The Transmission Selection Algorithm Table (12.21.2)
- c) The Priority Regeneration Override Table (12.21.3)

12.21.1 The Bandwidth Availability Parameter Table

There is one Bandwidth Availability Parameter Table per Port of a bridge component. Each table row contains a set of parameters for each traffic class that supports the credit-based shaper algorithm (8.6.8.2), as detailed in Table 12-1. Rows in the table can be created or removed dynamically in implementations that support dynamic configuration of ports and components.

Table 12-1—Bandwidth Availability Parameter Table row elements

Name	Data type	Operations supported*	Conformance†	References
Traffic class	unsigned integer [0..7]	R	BE	34.3
deltaBandwidth	percentage	RW	BE	34.3
adminIdleSlope	unsigned integer	RW	BE	34.3
operIdleSlope	unsigned integer	R	BE	34.3

*R = Read only access; RW = Read/Write access.

†B = Required for bridge or bridge component support of forwarding and queueing for time-sensitive streams.

E = Required for end station support of forwarding and queueing for time-sensitive streams.

12.21.2 The Transmission Selection Algorithm Table

There is one Transmission Selection Algorithm Table per Port of a bridge component. This is in addition to the Traffic Class Table managed object (12.6.3) that would also exist per Port of a bridge component. Each table row contains a set of parameters for each traffic class that the Port supports, as detailed in Table 12-2.

Table 12-2—Transmission Selection Algorithm Table row elements

Name	Data type	Operations supported [*]	Conformance [†]	References
Traffic class	unsigned integer [0..7]	R	B	8.6.8
Transmission selection algorithm	enumerated (see Table 8-4)	RW	B	8.6.8, Table 8-4

^{*}R = Read only access; RW = Read/Write access.

[†]B = Required for bridge or bridge component support of forwarding and queueing for time-sensitive streams.

12.21.3 The Priority Regeneration Override Table

There is one Priority Regeneration Override Table per Port of a bridge component. This is in addition to the Priority Handling managed object (12.6.2) that would also exist per Port of a bridge component. Each table row contains a set of parameters for each priority value that is associated with an SR class (3.3, 6.9.4), as detailed in Table 12-3.

Table 12-3—Priority Regeneration Override Table row elements

Name	Data type	Operations supported [*]	Conformance [†]	References
Received priority	integer [0..7]	R	B	6.9.4
Regenerated priority	integer [0..7]	RW	B	6.9.4
SRPdomainBoundaryPort	boolean	R	B	6.6.4

^{*}R = Read only access; RW = Read/Write access.

[†]B = Required for bridge or bridge component support of forwarding and queueing for time-sensitive streams.

17. Management protocol

17.2 Structure of the MIB

Insert a row at the appropriate position in Table 17-1, as shown:

Table 17-1—Structure of the MIB modules

Module	Subclause	Defining IEEE standard	Reference	Notes
IEEE8021-TC MIB	17.7.1	802.1ap	—	Textual conventions for all modules
IEEE8021-BRIDGE MIB	17.7.2	802.1D, 802.1p and 802.1t	802.1D	Adapted from IETF RFC 4188 and IETF RFC 4363
IEEE8021-SPANNING-TREE MIB	17.7.3	802.1w	802.1D 17	Adapted from IETF RFC 4318
IEEE8021-Q-BRIDGE MIB	17.7.4	802.1Q, 802.1u, and 802.1v	8	Adapted from IETF RFC 4363
IEEE8021-PB MIB	17.7.5	802.1ad	16	Initial version in IEEE Std 802.1ap
IEEE8021-MSTP MIB	17.7.6	802.1s	13	Initial version in IEEE Std 802.1ap
IEEE8021-CFM MIB	17.7.7.1	802.1ag	20	Initial version in IEEE Std 802.1ag
IEEE8021-CFM MIBV2	17.7.7.2	802.1ag	20	Initial version in IEEE Std 802.1ap
IEEE8021-PBB MIB	17.7.8	802.1ah	26	Initial version in IEEE Std 802.1ap
...	...	...	...	...
<u>IEEE8021-FQTSS MIB</u>	<u>17.7.13</u>	<u>802.1Qav</u>	<u>34</u>	<u>Initial version in IEEE Std 802.1Qav</u>

17.2.1 Structure of the IEEE8021-TC MIB

Insert the following NOTE after the initial paragraph:

NOTE—The IEEE8021-FQTSS MIB module in 17.7.13 defines additional TEXTUAL-CONVENTIONs that are utilized by the managed objects that support use of the Credit-based shaper algorithm (8.6.8.2, 34).

Insert new 17.2.13, and Table 17-18, as follows, renumbering Table 17-18 if necessary to follow in sequence from any tables referenced in the text prior to 17.2.13. Renumber subsequent tables/paragraphs as necessary:

17.2.13 Structure of the IEEE8021-FQTSS MIB

The IEEE8021-FQTSS MIB provides objects to configure and manage those aspects of a VLAN Bridge that are related to forwarding and queuing for time-sensitive streams (see Clause 34).

Objects in this MIB module are arranged into subtrees. Each subtree is organized as a set of related objects. Where appropriate, the corresponding Clause 12 management reference is also included.

Table 17-18 indicates the structure of the IEEE8021-FQTSS MIB module.

Table 17-18—FQTSS MIB structure and object cross reference

MIB table	MIB object	References
<i>ieee8021FqtssBap subtree</i>		
ieee8021FqtssBapTable		Bandwidth Availability Parameter Table, 12.21.1, 34.3
	ieee8021FqtssBAPTrafficClass	Traffic class (Table index)
	ieee8021FqtssDeltaBandwidth	deltaBandwidth, 12.21.1, 34.3
	ieee8021FqtssOperIdleSlopeMs	operIdleSlope, 12.21.1, 34.3
	ieee8021FqtssOperIdleSlopeLs	operIdleSlope, 12.21.1, 34.3
	ieee8021FqtssAdminIdleSlopeMs	adminIdleSlope, 12.21.1, 34.3
	ieee8021FqtssAdminIdleSlopeLs	adminIdleSlope, 12.21.1, 34.3
<i>ieee8021FqtssMappings subtree</i>		
ieee8021FqtssTxSelectionAlgorithmTable		Transmission Selection Algorithm Table, 12.21.2, 8.6.8
	ieee8021FqtssTrafficClass	Traffic class (Table index)
	ieee8021FqtssTxSelectionAlgorithmID	Transmission selection algorithm, 12.21.2, 8.6.8
ieee8021FqtssSrpRegenOverrideTable		SRP domain boundary port priority regeneration override table, 12.21.3, 6.6.4, 6.9.4
	ieee8021FqtssSrClassPriority	Received priority (Table index)
	ieee8021FqtssPriorityRegenOverride	Regenerated priority, 12.21.3, 6.9.4
	ieee8021FqtssSrpBoundaryPort	SRPdomainBoundaryPort, 12.21.3, 6.6.4

17.3 Relationship to other MIBs

Insert new 17.3.13, as follows, renumbering as necessary:

17.3.13 Relationship of the IEEE8021-FQTSS MIB to other MIB modules

The IEEE8021-FQTSS MIB provides objects that extend the core management functionality of a Bridge, as defined by the IEEE8021-BRIDGE-MIB (17.7.2), in order to support the additional management functionality needed when the forwarding and queuing for time-sensitive streams extensions, as defined in Clause 34, are supported by the Bridge. As support of the objects defined in the IEEE8021-FQTSS MIB also requires support of the IEEE8021-BRIDGE-MIB, the provisions of 17.3.2 apply to implementations claiming support of the IEEE8021-FQTSS MIB.

17.4 Security considerations

Insert new 17.4.13, as follows, renumbering as necessary:

17.4.13 Security considerations of the IEEE8021-FQTSS MIB

There are a number of management objects defined in the IEEE8021-FQTSS MIB module that have a MAX-ACCESS clause of read-write and/or read-create. Such objects may be considered sensitive or vulnerable in some network environments. The support for SET operations in a non-secure environment without proper protection can have a negative effect on network operations.

Some of the readable objects in this MIB module (i.e., objects with a MAX-ACCESS other than notaccessible) may be considered sensitive or vulnerable in some network environments. It is thus important to control all types of access (including GET and/or NOTIFY) to these objects and possibly to even encrypt the values of these objects when sending them over the network via SNMP.

The following tables and objects in the IEEE8021-FQTSS MIB can be manipulated to interfere with the operation of the forwarding and queuing mechanisms in a manner that would be detrimental to the transmission of time-sensitive streams:

```
ieee8021FqtssDeltaBandwidth
ieee8021FqtssAdminIdleSlopeMs
ieee8021FqtssAdminIdleSlopeLs
ieee8021FqtssTxSelectionAlgorithmID
ieee8021FqtssPriorityRegenOverride
```

- ieee8021FqtssDeltaBandwidth can be manipulated to reduce the amount of bandwidth available to a given SR class.
- ieee8021FqtssAdminIdleSlopeMs and ieee8021FqtssAdminIdleSlopeLs can be manipulated to change the overall amount of bandwidth available to stream traffic on a Port, in a network where SRP is not used for stream reservations.
- ieee8021FqtssTxSelectionAlgorithmID can be manipulated in order to apply the wrong transmission selection algorithm to a traffic class that was being used by stream traffic.
- ieee8021FqtssPriorityRegenOverride can be manipulated to disrupt stream traffic within an SRP domain with non-stream traffic entering from outside the SRP domain boundary.

17.7 MIB modules

Insert new 17.7.13, as follows, renumbering as necessary:

17.7.13 Definitions for the IEEE8021-FQTSS MIB module

```
IEEE8021-FQTSS-MIB DEFINITIONS ::= BEGIN

-- =====
-- MIB for support of 802.1Qav Forwarding & Queuing Enhancements
-- for Time Sensitive Streams (FQTSS) in 802.1Q Bridges.
-- =====

IMPORTS
    MODULE-IDENTITY,
    OBJECT-TYPE,
    Unsigned32
        FROM SNMPv2-SMI
    TEXTUAL-CONVENTION,
    TruthValue,
    RowStatus
```

```
FROM SNMPv2-TC
MODULE-COMPLIANCE,
OBJECT-GROUP
FROM SNMPv2-CONF
ieee802dot1mibs,
IEEE8021PriorityValue
FROM IEEE8021-TC-MIB
ieee8021BridgeBaseComponentId,
ieee8021BridgeBasePort
FROM IEEE8021-BRIDGE-MIB
;

ieee8021FqtssMib MODULE-IDENTITY
LAST-UPDATED "200910010000Z" -- October 1, 2009
ORGANIZATION "IEEE 802.1 Working Group"
CONTACT-INFO
    " WG-URL: http://grouper.ieee.org/groups/802/1/index.html
      WG-EMail: stds-802-1@ieee.org

      Contact: Tony Jeffree
      Postal: 11A POPLAR GROVE
              SALE, CHESHIRE
              M33 3AX
              UNITED KINGDOM
      Tel: +44-161-973-4278
      E-mail: tony@jeffree.co.uk"
DESCRIPTION
    "The Bridge MIB module for managing devices that support
    the IEEE 802.1Qav Forwarding and Queuing Enhancements
    for Time Sensitive Streams.

    Unless otherwise indicated, the references in this MIB
    module are to IEEE Std 802.1Q-2005 as amended by IEEE
    Std 802.1ad, IEEE Std 802.1ak, IEEE Std 802.1ag, IEEE
    Std 802.1ah, IEEE Std 802.1ap, and IEEE Std 802.1Qav.

    Copyright (C) IEEE.
    This version of this MIB module is part of IEEE802.1Q;
    see the draft itself for full legal notices."
REVISION    "200910010000Z" -- October 1, 2009
DESCRIPTION
    "Initial revision, included in IEEE 802.1Qav."
 ::= { ieee802dot1mibs 16 }

-- =====
-- Textual Conventions
-- =====

IEEE8021FqtssTrafficClassValue ::= TEXTUAL-CONVENTION
    DISPLAY-HINT "d"
    STATUS      current
    DESCRIPTION
        "An 802.1 FQTSS traffic class value.
        This is the numerical value associated with a traffic
        class in a Bridge. Larger values are associated with
        higher priority traffic classes."
    REFERENCE   "12.21"
    SYNTAX      Unsigned32 (0..7)
```

```

IEEE8021FqtssDeltaBandwidthValue ::= TEXTUAL-CONVENTION
    DISPLAY-HINT "d"
    STATUS      current
    DESCRIPTION
        "An 802.1 FQTSS delta bandwidth percentage,
        represented as a fixed point number scaled by
        1,000,000."
    REFERENCE   "12.21, 34.4"
    SYNTAX      Unsigned32 (0..100000000)

IEEE8021FqtssTxSelectionAlgorithmIDValue ::= TEXTUAL-CONVENTION
    DISPLAY-HINT "d"
    STATUS      current
    DESCRIPTION
        "An 802.1 transmission selection algorithm identifier
        value. This is an integer, with the following
        interpretation placed on the value:

        0: Strict priority algorithm,
        1: Credit-based shaper algorithm,
        2-255: Reserved for future standardization,
        256-4294967295: Vendor-specific transmission selection
            algorithm identifiers, consisting of a
            four-octet integer, where the 3 most
            significant octets hold an OUI value,
            and the least significant octet holds
            an integer value in the range 0-255
            assigned by the owner of the OUI."
    REFERENCE   "8.6.8, 12.21"
    SYNTAX      Unsigned32

-- =====
-- subtrees in the FQTSS MIB
-- =====

ieee8021FqtssNotifications
    OBJECT IDENTIFIER ::= { ieee8021FqtssMib 0 }

ieee8021FqtssObjects
    OBJECT IDENTIFIER ::= { ieee8021FqtssMib 1 }

ieee8021FqtssConformance
    OBJECT IDENTIFIER ::= { ieee8021FqtssMib 2 }

ieee8021FqtssBap
    OBJECT IDENTIFIER ::= { ieee8021FqtssObjects 1 }

ieee8021FqtssMappings
    OBJECT IDENTIFIER ::= { ieee8021FqtssObjects 2 }

-- =====
-- The ieee8021FqtssBap subtree
-- This subtree defines the objects necessary for the management
-- of bandwidth allocation for queues that support FQTSS.
-- =====

-- =====
-- the ieee8021FqtssBapTable
-- =====

```

ieee8021FqtssBapTable OBJECT-TYPE
 SYNTAX SEQUENCE OF Ieee8021FqtssBapEntry
 MAX-ACCESS not-accessible
 STATUS current
 DESCRIPTION
 "A table containing a set of bandwidth availability
 parameters for each traffic class that supports the
 credit-based shaper algorithm.
 All writable objects in this table must be
 persistent over power up restart/reboot."
 REFERENCE "12.21.1"
 ::= { ieee8021FqtssBap 1 }

ieee8021FqtssBapEntry OBJECT-TYPE
 SYNTAX Ieee8021FqtssBapEntry
 MAX-ACCESS not-accessible
 STATUS current
 DESCRIPTION
 "A list of objects containing bandwidth allocation
 information for each traffic class that supports the
 credit-based shaper algorithm. Rows in the table are
 automatically created and deleted as a result of the
 operation of the algorithm described in 34.5. "
 INDEX { ieee8021BridgeBaseComponentId,
 ieee8021BridgeBasePort,
 ieee8021FqtssBAPTrafficClass }
 ::= { ieee8021FqtssBapTable 1 }

Ieee8021FqtssBapEntry ::=

```

SEQUENCE {
    ieee8021FqtssBAPTrafficClass
        IEEE8021FqtssTrafficClassValue,
    ieee8021FqtssDeltaBandwidth
        IEEE8021FqtssDeltaBandwidthValue,
    ieee8021FqtssOperIdleSlopeMs
        Unsigned32,
    ieee8021FqtssOperIdleSlopeLs
        Unsigned32,
    ieee8021FqtssAdminIdleSlopeMs
        Unsigned32,
    ieee8021FqtssAdminIdleSlopeLs
        Unsigned32,
    ieee8021FqtssBapRowStatus
        RowStatus
}
```

ieee8021FqtssBAPTrafficClass OBJECT-TYPE
 SYNTAX IEEE8021FqtssTrafficClassValue
 MAX-ACCESS not-accessible
 STATUS current
 DESCRIPTION
 "The traffic class number associated with the row of
 the table.

 A row in this table is created for each traffic class
 that supports the credit-based shaper algorithm. The
 recommended mappings of priorities to traffic classes
 for support of the credit-based shaper algorithm are

described in 34.5.”
 REFERENCE “12.21.1, 34.3, 34.5”
 ::= { ieee8021FqtssBapEntry 1 }

ieee8021FqtssDeltaBandwidth OBJECT-TYPE

SYNTAX IEEE8021FqtssDeltaBandwidthValue
 UNITS “percent”
 MAX-ACCESS read-write
 STATUS current
 DESCRIPTION
 “The value of the deltaBandwidth parameter
 for the traffic class.
 This value is represented as a fixed point number
 scaled by a factor of 1,000,000; i.e., 100,000,000
 (the maximum value) represents 100%.

The default value of the deltaBandwidth parameter
 for the highest numbered traffic class that supports
 the credit-based shaper algorithm is 75%; for all
 lower numbered traffic classes that support the
 credit-based shaper algorithm the default value is 0%.

The value of this object MUST be retained across
 reinitializations of the management system.”
 REFERENCE “12.21.1, 34.3”
 ::= { ieee8021FqtssBapEntry 2 }

ieee8021FqtssOperIdleSlopeMs OBJECT-TYPE

SYNTAX Unsigned32
 UNITS “bits per second”
 MAX-ACCESS read-only
 STATUS current
 DESCRIPTION
 “The most significant 32 bits of the bandwidth,
 in bits per second, that is currently allocated to the
 traffic class (idleSlope(N)). This object MUST be read
 at the same time as ieee8021FqtssOperIdleSlopeLs,
 which represents the LS 32 bits of the value, in order
 for the read operation to succeed.

If SRP is supported and in operation, then the reserved
 bandwidth is determined by the operation of SRP; otherwise,
 the value of ieee8021FqtssOperIdleSlopeMs is equal to
 the value of ieee8021FqtssAdminIdleSlopeMs.

The value of this object MUST be retained across
 reinitializations of the management system.”
 REFERENCE “12.21.1, 34.3”
 ::= { ieee8021FqtssBapEntry 3 }

ieee8021FqtssOperIdleSlopeLs OBJECT-TYPE

SYNTAX Unsigned32
 UNITS “bits per second”
 MAX-ACCESS read-only
 STATUS current
 DESCRIPTION
 “The least significant 32 bits of the bandwidth,
 in bits per second, that is currently allocated to the

traffic class (idleSlope(N)). This object MUST be read at the same time as ieee8021FqtssOperIdleSlopeMs, which represents the LS 32 bits of the value, in order for the read operation to succeed.

If SRP is supported and in operation, then the reserved bandwidth is determined by the operation of SRP; otherwise, the value of ieee8021FqtssOperIdleSlopeLs is equal to the value of ieee8021FqtssAdminIdleSlopeMs.

The value of this object MUST be retained across reinitializations of the management system."

REFERENCE "12.21.1, 34.3"
::= { ieee8021FqtssBapEntry 4 }

ieee8021FqtssAdminIdleSlopeMs OBJECT-TYPE

SYNTAX Unsigned32
UNITS "bits per second"
MAX-ACCESS read-write
STATUS current

DESCRIPTION

"The most significant 32 bits of the bandwidth, in bits per second, that the manager desires to allocate to the traffic class as idleSlope(N). This object MUST be read or written at the same time as ieee8021FqtssAdminIdleSlopeLs, which represents the LS 32 bits of the value, in order for the read or write operation to succeed.

If SRP is supported and in operation, then the reserved bandwidth is determined by the operation of SRP, and any changes to the value of this object have no effect on the operational value of idleSlope(N).

The value of this object MUST be retained across reinitializations of the management system."

REFERENCE "12.21.1, 34.3"
DEFVAL { 0 }
::= { ieee8021FqtssBapEntry 5 }

ieee8021FqtssAdminIdleSlopeLs OBJECT-TYPE

SYNTAX Unsigned32
UNITS "bits per second"
MAX-ACCESS read-write
STATUS current

DESCRIPTION

"The least significant 32 bits of the bandwidth, in bits per second, that the manager desires to allocate to the traffic class as idleSlope(N). This object MUST be read or written at the same time as ieee8021FqtssAdminIdleSlopeMs, which represents the LS 32 bits of the value, in order for the read or write operation to succeed.

If SRP is supported and in operation, then the reserved bandwidth is determined by the operation of SRP, and any changes to the value of this object have no effect on the operational value of idleSlope(N).

The value of this object MUST be retained across reinitializations of the management system."

REFERENCE "12.21.1, 34.3"

DEFVAL { 0 }

::= { ieee8021FqtssBapEntry 6 }

ieee8021FqtssBapRowStatus OBJECT-TYPE

SYNTAX RowStatus

MAX-ACCESS read-create

STATUS current

DESCRIPTION

"Indicates the status of an entry (row) in this table, and is used to create/delete entries.

The corresponding instances of the following objects must be set before this object can be made active(1):

ieee8021FqtssBAPTrafficClass

ieee8021FqtssDeltaBandwidth

ieee8021FqtssOperIdleSlopeMs

ieee8021FqtssOperIdleSlopeLs

ieee8021FqtssAdminIdleSlopeMs

ieee8021FqtssAdminIdleSlopeLs

The corresponding instances of the following objects may not be changed while this object is active(1):

ieee8021FqtssBAPTrafficClass"

::= { ieee8021FqtssBapEntry 7 }

```
-- =====
-- The ieee8021FqtssMappings subtree
-- This subtree defines the objects necessary for the assignment
-- of transmission selection algorithms to traffic classes,
-- and definition of regeneration table override values.
-- =====

-- =====
-- the ieee8021FqtssTxSelectionAlgorithmTable
-- =====
```

ieee8021FqtssTxSelectionAlgorithmTable OBJECT-TYPE

SYNTAX SEQUENCE OF Ieee8021FqtssTxSelectionAlgorithmEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"A table containing the assignment of transmission selection algorithms to traffic classes for the Port. This table provides management of the Transmission Selection Algorithm Table defined in 8.6.8.

For a given Port, a row in the table exists for each traffic class that is supported by the Port.

The default assignments of transmission selection algorithms to traffic classes in the table are made on instantiation of the table, in accordance with the defaults defined in 8.6.8 and 34.5.

All writable objects in this table must be persistent over power up restart/reboot."

REFERENCE "8.6.8, 12.21.2, 34.5"
 ::= { ieee8021FqtssMappings 1 }

ieee8021FqtssTxSelectionAlgorithmEntry OBJECT-TYPE
 SYNTAX Ieee8021FqtssTxSelectionAlgorithmEntry
 MAX-ACCESS not-accessible
 STATUS current
 DESCRIPTION
 "A list of objects that contain the mapping of a
 traffic class value to a transmission selection algorithm
 value."
 INDEX { ieee8021BridgeBaseComponentId,
 ieee8021BridgeBasePort,
 ieee8021FqtssTrafficClass }
 ::= { ieee8021FqtssTxSelectionAlgorithmTable 1 }

Ieee8021FqtssTxSelectionAlgorithmEntry ::=
 SEQUENCE {
 ieee8021FqtssTrafficClass
 IEEE8021FqtssTrafficClassValue,
 ieee8021FqtssTxSelectionAlgorithmID
 IEEE8021FqtssTxSelectionAlgorithmIDValue
 }

ieee8021FqtssTrafficClass OBJECT-TYPE
 SYNTAX IEEE8021FqtssTrafficClassValue
 MAX-ACCESS not-accessible
 STATUS current
 DESCRIPTION
 "The traffic class to which the transmission selection
 algorithm is assigned.

 The value of this object MUST be retained across
 reinitializations of the management system."
 REFERENCE "8.6.8, 12.21.2, 34.5"
 ::= { ieee8021FqtssTxSelectionAlgorithmEntry 1 }

ieee8021FqtssTxSelectionAlgorithmID OBJECT-TYPE
 SYNTAX IEEE8021FqtssTxSelectionAlgorithmIDValue
 MAX-ACCESS read-write
 STATUS current
 DESCRIPTION
 "The identifier of the transmission selection algorithm
 assigned to the traffic class.

 The value of this object MUST be retained across
 reinitializations of the management system."
 REFERENCE "8.6.8, 12.21.2, 34.5"
 ::= { ieee8021FqtssTxSelectionAlgorithmEntry 2 }

-- =====
 -- the ieee8021FqtssSrpRegenOverrideTable
 -- =====

ieee8021FqtssSrpRegenOverrideTable OBJECT-TYPE
 SYNTAX SEQUENCE OF Ieee8021FqtssSrpRegenOverrideEntry
 MAX-ACCESS not-accessible
 STATUS current

DESCRIPTION

"A table containing the set of priority regeneration table override values for the Port.

The recommended default values of priorities associated with SR classes, and the corresponding override values, are defined in 6.9.4.

All writable objects in this table must be persistent over power up restart/reboot."

REFERENCE "12.21.3, 6.6.4, 6.9.4"

::= { ieee8021FqtssMappings 2 }

ieee8021FqtssSrpRegenOverrideEntry OBJECT-TYPE

SYNTAX Ieee8021FqtssSrpRegenOverrideEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"A list of objects that contain the mapping of a priority value to a priority regeneration override value, and a boundary port indication.

Rows in the table exist for all priorities that are associated with SR classes."

INDEX { ieee8021BridgeBaseComponentId,
ieee8021BridgeBasePort,
ieee8021FqtssSrClassPriority }

::= { ieee8021FqtssSrpRegenOverrideTable 1 }

Ieee8021FqtssSrpRegenOverrideEntry ::=

```
SEQUENCE {
    ieee8021FqtssSrClassPriority
        IEEE8021PriorityValue,
    ieee8021FqtssPriorityRegenOverride
        IEEE8021PriorityValue,
    ieee8021FqtssSrpBoundaryPort
        TruthValue
}
```

ieee8021FqtssSrClassPriority OBJECT-TYPE

SYNTAX IEEE8021PriorityValue

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"The priority value that is overridden at the SRP domain boundary. "

REFERENCE "12.21.3, 6.6.4, 6.9.4"

::= { ieee8021FqtssSrpRegenOverrideEntry 1 }

ieee8021FqtssPriorityRegenOverride OBJECT-TYPE

SYNTAX IEEE8021PriorityValue

MAX-ACCESS read-write

STATUS current

DESCRIPTION

"The priority value that is used to override the priority regeneration table entry at the SRP domain boundary.

The value of this object MUST be retained across reinitializations of the management system."

```

REFERENCE    "12.21.3, 6.6.4, 6.9.4"
::= { ieee8021FqtssSrpRegenOverrideEntry 2 }

ieee8021FqtssSrpBoundaryPort OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value of the SRPdomainBoundaryPort parameter
        (6.6.4) for the priority. "
    REFERENCE   "12.21.3, 6.6.4, 6.9.4"
    ::= { ieee8021FqtssSrpRegenOverrideEntry 3 }

-- =====
-- IEEE8021 FQTSS MIB - Conformance Information
-- =====

ieee8021FqtssCompliances
    OBJECT IDENTIFIER ::= { ieee8021FqtssConformance 1 }
ieee8021FqtssGroups
    OBJECT IDENTIFIER ::= { ieee8021FqtssConformance 2 }

-- =====
-- units of conformance
-- =====

-- =====
-- the ieee8021FqtssBap group
-- =====

ieee8021FqtssBapGroup OBJECT-GROUP
    OBJECTS {
        ieee8021FqtssDeltaBandwidth,
        ieee8021FqtssOperIdleSlopeMs,
        ieee8021FqtssOperIdleSlopeLs,
        ieee8021FqtssAdminIdleSlopeMs,
        ieee8021FqtssAdminIdleSlopeLs,
        ieee8021FqtssBapRowStatus
    }
    STATUS      current
    DESCRIPTION
        "Objects that define bandwidth allocation for FQTSS."
    ::= { ieee8021FqtssGroups 1 }

-- =====
-- the ieee8021FqtssTxSelectionAlgorithm group
-- =====

ieee8021FqtssTxSelectionAlgorithmGroup OBJECT-GROUP
    OBJECTS {
        ieee8021FqtssTxSelectionAlgorithmID
    }
    STATUS      current
    DESCRIPTION
        "Objects that define transmission selection
        mappings for FQTSS."
    ::= { ieee8021FqtssGroups 2 }

```

```

-- =====
-- the ieee8021FqtssBoundaryPort group
-- =====

ieee8021FqtssBoundaryPortGroup OBJECT-GROUP
    OBJECTS {
        ieee8021FqtssPriorityRegenOverride,
        ieee8021FqtssSrpBoundaryPort
    }
    STATUS      current
    DESCRIPTION
        "Objects that define boundary port priority override
        mappings for FQTSS."
    ::= { ieee8021FqtssGroups 3 }

-- =====
-- compliance statements
-- =====

ieee8021FqtssCompliance MODULE-COMPLIANCE
    STATUS      current
    DESCRIPTION
        "The compliance statement for devices supporting
        forwarding and queuing for time sensitive streams.

        Support of the objects defined in the IEEE8021-FQTSS MIB
        also requires support of the IEEE8021-BRIDGE-MIB; the
        provisions of 17.3.2 apply to implementations claiming
        support of the IEEE8021-FQTSS MIB. "

    MODULE -- this module
        MANDATORY-GROUPS {
            ieee8021FqtssBapGroup,
            ieee8021FqtssTxSelectionAlgorithmGroup,
            ieee8021FqtssBoundaryPortGroup
        }

    ::= { ieee8021FqtssCompliances 1 }

END

```


Insert the following text, Table 34-1, and Figure 34-1, as new Clause 34:

34. Forwarding and queuing for time-sensitive streams

34.1 Overview

This clause describes a set of tools that can be used to support the forwarding and queuing requirements of time-sensitive streams. In this context, a “time-sensitive stream” is taken to be a stream of traffic, transmitted from a single source station, destined for one or more destination stations, where the traffic is sensitive to timely delivery, and in particular, requires transmission latency to be bounded. Such streams include video or audio data streams, where there is a desire to limit the amount of buffering required in the receiving station.

NOTE 1—An example of this requirement would be a live performance where a video image of the performance is simultaneously projected on a screen in the auditorium, and it is desirable for the sound and image to be “in synch” with the performance.

In order to address the needs of such traffic in Bridges, the following are provided:

- a) A means of detecting the boundary between a set of Bridges that support SRP (an SRP domain) and surrounding Bridges that do not support SRP. This mechanism is described in 34.2.

NOTE 2—The primary intent of these functions is to support SRP; however, there is no specific interdependency between these functions and SRP, so they could equally be used to support other admission control mechanisms if they were implemented.

- b) A set of bandwidth availability parameters for each port that are used to record the maximum bandwidth available to a given outbound queue, and the actual bandwidth reserved, for that queue. These parameters are described in 34.3.
- c) A credit-based shaper algorithm, defined in 8.6.8.1, that is used to shape the transmission of stream-based traffic in accordance with the bandwidth that has been reserved on a given outbound queue.
- d) A discussion of the relationship between the size of the layer 2 “payload” (the MSDU) carried in a frame and how that relates to the actual bandwidth that will be consumed when that MSDU is transmitted on a particular Port (34.4).
- e) An algorithm for determining the mapping of the priorities associated with received frames onto the traffic classes available on the transmission Ports of a Bridge (34.5).
- f) A definition of the required behavior of an end station that acts as the source of a time-sensitive data stream (34.6).

34.2 Detection of SRP domains

The concept of AV streams, the Stream Reservation Protocol (SRP), and the traffic forwarding and shaping functions that support stream transmission (see 6.9.4 and 8.6.8.1), rely on the ability of each bridge to detect whether each of its ports is at the edge of a region of connected bridges that support SRP on a particular priority, so that the priority code point (PCP) values associated with traffic entering an *SRP domain* (3.4) can be properly regenerated at the boundary of the domain, as described in 6.9.4.

Bridges detect the edge of an SRP domain by observing SRP behavior. If a Bridge receives SRP registrations using a particular priority, then it is reasonable to believe that they are being received from an SRP capable device; the SRP engine can therefore signal which Ports of a Bridge are at the boundary of an SRP domain. The per-port, per-SR class, *SRPdomainBoundaryPort* parameter determines whether or not a

Port is considered to be at the edge of an SRP domain or within the core of the domain, as defined in 6.6.4. This parameter is controlled by the operation of SRP.

NOTE 1—SRP domain detection is based on the assumption that a device connected to a Port is either SRP capable for a given SR class, or is not SRP capable for that SR class. The detection is based on current reservation activity; the boundary of a domain will therefore expand to include Ports as SRP attributes are declared. The position of the domain boundary has no effect on the transmission of SRP frames; rather, it reflects where SRP activity is occurring. Ports are removed from the SRP domain when they are removed from the active topology.

34.3 The bandwidth availability parameters

The following bandwidth availability parameters exist for each Port, and for each traffic class, N , that supports the credit-based shaper algorithm:

- a) ***portTransmitRate*** as defined in 8.6.8.2;
- b) ***deltaBandwidth(N)***: The additional bandwidth, represented as a percentage of *portTransmitRate*, that can be reserved for use by the queue associated with traffic class N , in addition to the *deltaBandwidth(N)* values associated with higher priority queues. For a given traffic class N , the total bandwidth that can be reserved is the sum of the *deltaBandwidth* values for traffic class N and all higher traffic classes, minus any bandwidth reserved by higher traffic classes that support the credit-based shaper algorithm (see 34.3.1).
- c) ***adminIdleSlope(N)***: The bandwidth, in bits per second, that has been requested by management to be reserved for use by the queue associated with traffic class N . If SRP is in operation, this parameter has no effect; if SRP is not in operation, then the value of *operIdleSlope(N)* is always equal to the value of *adminIdleSlope(N)*.
- d) ***operIdleSlope(N)***: The actual bandwidth, in bits per second, that is currently reserved for use by the queue associated with traffic class N . This value is used by the credit-based shaper algorithm (8.6.8.2) as the *idleSlope* for the corresponding queue.

In all cases, bandwidth is defined in terms of the actual bandwidth consumed when frames are transmitted through the Port, not the size of the MSDU “payload” carried within those frames. 34.4 discusses the relationship between MSDU size and actual bandwidth consumed.

NOTE—While the *deltaBandwidth* values are specified with respect to specific traffic classes, and indicate the amount of bandwidth that can be reserved for traffic belonging to a particular traffic class, this does not imply that these traffic classes have preferential access to that portion of the bandwidth. The priority of a given traffic class does not, for example, imply anything about the importance of a data stream that uses that class; the reservation strategy might therefore allocate bandwidth to a high importance stream that uses a lower priority traffic class in preference to a stream of lower importance that uses a higher priority traffic class.

34.3.1 Relationships among bandwidth availability parameters

The recommended default value of *deltaBandwidth(N)* for the highest numbered traffic class supported is 75%, and for any lower numbered traffic classes, the recommended default value is 0%. The *deltaBandwidth(N)* for a given N , plus the *deltaBandwidth(N)* values for any higher priority queues (larger values of N) defines the total percentage of the Port’s bandwidth that can be reserved for that queue and all higher priority queues. For the highest priority queue, this means that the maximum value of *operIdleSlope(N)* is *deltaBandwidth(N)*% of *portTransmitRate*. However, if *operIdleSlope(N)* is actually less than this maximum value, any lower priority queue that supports the credit-based shaper algorithm can make use of the reservable bandwidth that is unused by the higher priority queue. So, for queue $N-1$, the maximum value of (*operIdleSlope(N)* + *operIdleSlope(N-1)*) is (*deltaBandwidth(N)* + *deltaBandwidth(N-1)*)% of *portTransmitRate*.

NOTE 1—For example, suppose two queues, 3 and 2, support the credit-based shaper algorithm for SR classes A and B respectively. Suppose *deltaBandwidth(3)* for SR class A is currently 20%, and *deltaBandwidth(2)* for SR class B is

currently 30%. If *operIdleSlope(3)* is currently 10% of *portTransmitRate*, then half of queue 3's maximum allocation is unused, and the maximum value of *operIdleSlope(2)* is therefore currently 40% of *portTransmitRate*. However, if *operIdleSlope(3)* increases to the full 20% that it is entitled to use, the maximum value of *operIdleSlope(2)* reduces to 30% of *portTransmitRate*.

NOTE 2—The sum of the *deltaBandwidth(N)* values for all values of N should be chosen such that there is sufficient bandwidth available for any non-reserved (best-effort, strict-priority) traffic; the default values are chosen such that the sum of the *deltaBandwidth(N)* values is 75%, so no more than 75% of the Port's available bandwidth is permitted to be reserved. This ensures that when using default settings, there is at least 25% of the Port's bandwidth available for non-reserved traffic. However, as these default settings may be inappropriate for some situations (e.g., links that offer very high bandwidth, or networks with very low levels of non-reserved traffic), they can be modified by management.

34.3.2 Bandwidth availability parameter management

The values of *deltaBandwidth(N)* and *adminIdleSlope(N)* can be changed by management, using the management operations defined in 12.21. If the stream reservation mechanisms defined in SRP are supported, then the values of *operIdleSlope(N)* are determined solely by the operation of SRP. If the stream reservation mechanisms defined in SRP are not supported, then the values of *operIdleSlope(N)* are equal to the values requested by management in the corresponding *adminIdleSlope(N)* parameters.

It is possible for the value of *portTransmitRate* for a Port to change as a result of the normal operation of the underlying MAC service (e.g., as a result of the operation of IEEE Std 802.1AX link aggregation, or as a result of dynamic changes in bandwidth of the physical layer technology itself, as can occur in wireless LAN technologies); it is also possible for management action to change the values of *deltaBandwidth(N)* for a Port. In either case, the consequence could be one of the following:

- a) The sum of the *operIdleSlope(N)* values for the Port could now exceed the total reservable bandwidth allowed for the Port, or the *operIdleSlope(N)* value for a given queue could now exceed the reservable bandwidth allowed for the queue, as defined in 34.3.1. Consequently, there could be streams currently active on the Port that can no longer be supported.
- b) The bandwidth now available to a given queue could mean that there are streams that are currently inactive that could be supported on the Port.
- c) Active streams that continue to be supported after the change could see their latency guarantee change.

In either case, corrective action, either by management or by the stream reservation mechanisms defined in SRP, is required in order to restore the parameters to a consistent set of values. In order to indicate that there have been changes in the bandwidth availability database, a signal, *bandwidthAvailabilityChanged*, is generated for each Port whenever the bandwidth availability parameters *portTransmitRate* or *deltaBandwidth(N)* are changed; this signal is used by SRP to trigger recalculation of the set of streams that can be reserved on the Port.

NOTE—During recalculation of stream reservations, the Bridge might be temporarily unable to honor bandwidth reservation commitments or forward best-effort traffic.

34.4 Deriving actual bandwidth requirements from the size of the MSDU

The forwarding and queuing mechanisms defined in this clause use bandwidth parameters that are defined in terms of the actual bandwidth used when frames are transmitted on the medium that supports the MAC service available through the Port. In contrast, the Stream Reservation Protocol (SRP) makes use of a traffic specification (TSpec) for each stream that defines the maximum number of bits per frame (*MaxFrameSize*), of the *mac_service_data_unit* parameter that is relayed by the relay function of the Bridge, and a maximum frame rate (*MaxIntervalFrames*), in frames per class measurement interval, for that stream; i.e., the TSpec takes no account of the per-frame overhead associated with transmitting the MSDU over a given medium. However, when SRP determines the value to be used for the *operIdleSlope(N)* parameter associated with a

given queue, it is necessary for this value to include the per-frame overhead that will be incurred when frames are transmitted on that Port.

NOTE 1—The frame rate in a TSpec is measured over a “class measurement interval” that depends upon the SR class associated with the stream. SR class A corresponds to a class measurement interval of 125 μ s; SR class B corresponds to a class measurement interval of 250 μ s. These class measurement intervals apply at the source of the stream, i.e., the “talker” end station, and do not necessarily hold good for subsequent stages in the stream’s transmission across a Bridged LAN.

For the purposes of calculating the bandwidth consumption of a stream, it is assumed that the stream data is essentially of constant size and transmission rate, so these maxima can be used to directly define an assumed maximum payload size and the maximum frame rate in frames per second; i.e.,:

$$\text{assumedPayloadSize} = \text{MaxFrameSize} \quad (1)$$

$$\text{maxFrameRate} = \text{MaxIntervalFrames} \times (1/\text{classMeasurementInterval}) \quad (2)$$

where *classMeasurementInterval* is measured in seconds.

NOTE 2—As stated, the calculation of bandwidth from TSpec parameters assumes that the stream data is essentially of constant frame size, and hence, the approximations shown in this section are valid. If the data varies significantly in frame size, then the calculation of per-frame overhead using these assumptions could be significantly in error.

From this, and also from local knowledge of the protocol stack that supports the Bridge Port, it is possible to determine the overhead that is added to the per-frame MSDU payload when a frame is transmitted. There are at least the following sources of per-frame overhead:

- a) Any VLAN tags and security tags (see IEEE Std 802.1AE) that are added to the layer 2 payload as it passes through the various service interfaces in the Port’s protocol stack.
- b) The MAC framing (header and trailer octets, plus any padding octets that are required to meet minimum frame size limitations) that is added by the underlying MAC service.
- c) Any physical layer overhead, such as preamble characters and inter-frame gaps.

The precise per-frame overhead will therefore depend upon the protocol stack and the underlying MAC technology.

The actual bandwidth needed to support a given stream is therefore defined as follows [using *assumedPayloadSize* from Equation (1)]:

$$\text{actualBandwidth} = (\text{perFrameOverhead} + \text{assumedPayloadSize}) \times \text{maxFrameRate} \quad (3)$$

34.5 Mapping priorities to traffic classes for time-sensitive streams

In Bridges that support forwarding and queuing for time-sensitive streams, the default mappings of priorities to traffic classes meet the following constraints:

- a) Priority values that correspond to SR classes are mapped onto traffic classes that support the credit-based shaper algorithm as the transmission selection algorithm.
- b) Traffic classes that support the credit-based shaper algorithm have a higher priority than traffic classes that support the strict priority (or any other) transmission selection algorithm.
- c) At least one traffic class supports the credit-based shaper algorithm, and at least one traffic class supports the strict priority transmission selection algorithm.

NOTE 1—The constraint that there is at least one traffic class that supports the strict priority transmission selection ensures that there is at least one traffic class that can support traffic that is not subject to bandwidth reservation, such as “best effort” traffic.

The recommended default priority to traffic class mappings for a system that supports SR class A (using priority 3) and SR class B (using priority 2) are shown in Table 34-1. The recommended default priority to traffic class mappings for a system that supports only SR class B (using priority 2) are shown in Table 34-2. The corresponding default configuration for the Transmission Selection Algorithm Table (see 8.6.8) is that the traffic classes that are shaded in the tables are configured to use the credit-based shaper algorithm, and the remaining traffic classes are configured to use the strict priority algorithm.

Table 34-1—Recommended priority to traffic class mappings for SR classes A and B

		Number of Available Traffic Classes						
		2	3	4	5	6	7	8
Priority	0 (Default)	0	0	0	0	0	0	1
	1	0	0	0	0	0	0	0
	2	1	1	2	3	4	5	6
	3	1	2	3	4	5	6	7
	4	0	0	1	1	1	1	2
	5	0	0	1	1	1	2	3
	6	0	0	1	2	2	3	4
	7	0	0	1	2	3	4	5

Table 34-2—Recommended priority to traffic class mappings for SR class B only

		Number of Available Traffic Classes						
		2	3	4	5	6	7	8
Priority	0 (Default)	0	0	0	0	0	1	1
	1	0	0	0	0	0	0	0
	2	1	2	3	4	5	6	7
	3	0	0	0	1	1	2	2
	4	0	1	1	2	2	3	3
	5	0	1	1	2	2	3	4
	6	0	1	2	3	3	4	5
	7	0	1	2	3	4	5	6

NOTE 2—The mapping shown in Table 34-1 for the case of only two available traffic classes maps two SR classes to a single traffic class. While this is a permissible configuration, in general it is desirable, and in some applications, can be a requirement, to assign SR classes to distinct traffic classes, as is done in this table for the cases where three or more traffic classes are available. The mappings shown deal only with one or two supported SR classes; a similar mapping strategy can be adopted if more than two SR classes are supported.

34.6 End station behavior

In order for an end station to successfully participate in the transmission and reception of time-sensitive streams, it is necessary for their behavior to be compatible with the operation of the forwarding and queueing mechanisms employed in bridges. The requirements for end stations that participate as “talkers”—i.e., sources of time-sensitive streams—are different from the requirements that apply to “listeners”—i.e., the destination station(s) for the streams.

34.6.1 Talker behavior

In order for Talker-originated data streams to make use of the credit-based shaper behavior in Bridges, it is a requirement for a Talker to use the priorities that the Bridges in the network recognize as being associated with SR classes exclusively for transmitting stream data. It is also necessary for the Talker, and the Bridges in the path to the Listener(s), to have a common view of the bandwidth required in order to transmit the Talker’s streams, and for that bandwidth to be reserved along the path to the Listener(s). This latter requirement can be met by means of stream reservation mechanisms, such as defined in SRP, or by other management means.

End stations that are Talkers shall exhibit transmission behavior for frames that are part of time-sensitive streams that is consistent with the operation of the credit-based shaper algorithm, both in terms of the way they transmit frames that are part of an individual data stream, and in terms of the way they transmit stream data frames from a Port. In effect, the queuing model for a Talker Port, and for a given priority, can be considered to look like Figure 34-1.

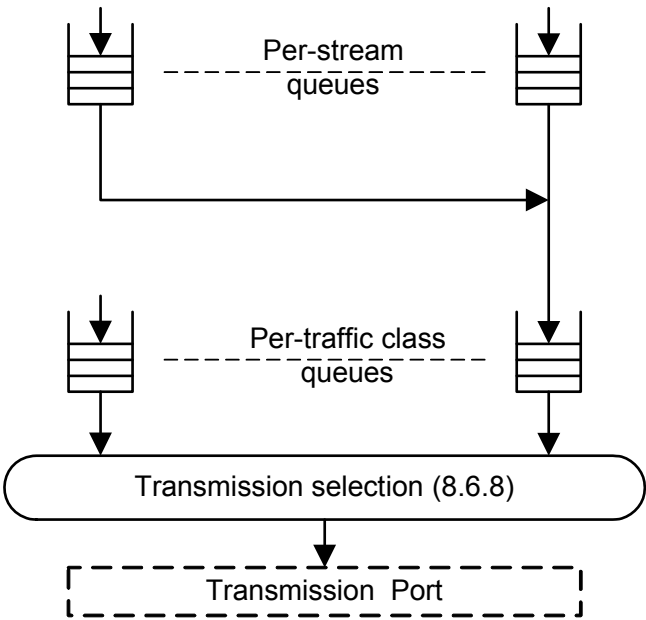


Figure 34-1—Queuing model for a Talker station

The Talker places frames into the queue associated with an individual stream based on the *Tspec* for that stream; i.e., during each class measurement interval, it can place up to *MaxIntervalFrames* data frames, each no longer than *MaxFrameSize* into that stream's queue.

The queue associated with each individual stream uses the credit-based shaper algorithm, with the *idleSlope* set to the bandwidth requirement of the stream on that transmission Port, as the means of determining the rate at which data frames for that stream are placed in the outbound queue for the priority that the stream is using. The outbound queue for that priority, in turn, makes use of the credit-based shaper algorithm, with the *idleSlope* set to the sum of the *idleSlope* values for all streams using that priority on that transmission Port, as the means of determining the rate at which data frames for that stream are selected for transmission.

Transmission selection in a Talker operates in the same way as in a Bridge; from this point of view, a Talker can be thought of as if it is a single-port Bridge.

For streams that make use of SR class A or SR class B, it is a requirement that the rate at which frames for any given stream are selected for placement in its per-stream queue does not exceed the bandwidth reserved for the stream, measured over the class measurement interval for the SR class (125 μ s for SR class A, 250 μ s for SR class B.) For some combinations of stream bandwidth requirement and transmission Port data rate, this can place a limit on the frame size that can be used when transmitting stream data.

NOTE—the sole implication of the observation interval is its effect on frame size, because the shaper behavior itself is independent of observation interval. The intent in limiting the observation interval is to limit frame size, as this is a major contribution to the latency experienced by a frame in transit through an AVB network (see discussion in Annex L).

34.6.2 Listener behavior

The primary requirement for a listener station is that it is capable of buffering the amount of data that could be transmitted for a stream during a time period equivalent to the accumulated maximum jitter that could be experienced by stream data frames in transmission between Talker and Listener. From the point of view of the specification of the forwarding and queuing requirements for time-sensitive streams, it is assumed that the listener will assess the buffering required for a stream as part of the stream bandwidth reservation mechanisms employed by the implementation.

Annex A

(normative)

PICS proforma—Bridge implementations³

A.5 Major capabilities

Change the first row of Table A.5 as shown:

	If the implementation is an end station implementation, mark “N/A” and continue at Annex I .			N/A []
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Insert the following row at the end of Table A.5:

FQTSS	Does the implementation support forwarding and queuing for time-sensitive streams?	O	5.4.1.5, 34	Yes []	No []
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A.14 Bridge management

Insert the following row at the end of Table A.14, re-numbering item MGT-97 if necessary:

Item	Feature	Status	References	Support
MGT-97	Does the implementation support the management entities defined in 12.21?	FQTSS: O	5.4.1.5 item e), 12.21, 17.7.13, 34	Yes [] N/A []

A.24 Management Information Base (MIB)

Insert the following row at the end of Table A.24, re-numbering item MIB-21 if necessary:

Item	Feature	Status	References	Support
MIB-21	Is the IEEE8021-FQTSS-MIB module fully supported (per its MODULE-COMPLIANCE)?	FQTSS: O	5.4.1.5 item e), 12.21, 17.7.13, 34	Yes [] N/A []

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Insert new Table A.31 as shown:

A.31 Forwarding and queuing for time-sensitive streams

Item	Feature	Status	References	Support
	If forwarding and queuing for time-sensitive streams (FQTSS in Table A.31) is not supported, mark N/A and ignore the remainder of this table.		5.4.1.5, 34	N/A []
FQTSSE1	Support a minimum of two traffic classes, of which one supports the strict priority algorithm and the other is an SR class.	FQTSS:M	5.4.1.5, 8.6.8.1, 34	Yes [] N/A []
FQTSSE2	Support the operation of the credit-based shaper algorithm on all Ports as the transmission selection algorithm used for the SR class.	FQTSS:M	5.4.1.5, 8.6.8.2, 34	Yes [] N/A []
FQTSSE3	Support SRP domain boundary port priority regeneration override as defined in 6.9.4, and the default priority regeneration override value defined in Table 6-6, for SR class “B”.	FQTSS:M	5.4.1.5, 6.9.4, Table 6-6, 34	Yes [] N/A []
FQTSSE4	Support the tables and procedures for mapping priorities to traffic classes as defined in 34.5.	FQTSS:M	5.4.1.5, 34, 34.5	Yes [] N/A []
FQTSSE5	Support two or more SR classes (a maximum of seven), and support the operation of the credit-based shaper algorithm on all Ports as the transmission selection algorithm used for those SR classes. The number of SR classes supported shall be stated in the PICS.	FQTSS:O	5.4.1.5, 8.6.8.2, 34.6	Yes [] No [] Number _____
FQTSSE6	Support SRP domain boundary port priority regeneration override as defined in 6.9.4, and the default priority regeneration override value defined in Table 6-6, for SR class “A”. If more than two SR classes are supported, the default priority regeneration override values used for the additional SR classes shall be stated in the PICS.	FQTSS:O	5.4.1.5, 6.9.4, Table 6-6, 34.6	Yes [] No [] Priority override values _____

Annex H

(informative)

Bibliography

Insert the following bibliography entries at the end of the existing list, renumbering if necessary:

[B29] Calculating the Delay Added by Qav Stream Queue (see <http://www.ieee802.org/1/files/public/docs2009/av-fuller-queue-delay-calculation-0809-v02.pdf>).

[B30] IEEE P802.1AS/D6.1, Draft Standard for Local and Metropolitan Area Networks—Timing and Synchronization for Time-Sensitive Applications in Bridged Local Area Networks.

[B31] IEEE P802.1Qat/D4.2, Draft Standard for Local and Metropolitan Area Networks—Virtual Bridged Local Area Networks—Amendment XX: Stream Reservation Protocol (SRP).

Annex I

(normative)

PICS proforma—End station implementations⁴

Change Table I.5 as follows:

I.5 Major capabilities

Item	Feature	Status	References	Support	
<u>MRPAP</u>	<u>Does the implementation support any MRP applications? If “No” is marked, continue at FQTSSE</u>	<u>O</u>	<u>5.12.1</u>	<u>Yes []</u>	<u>No []</u>
MMRP	Is the operation of MMRP supported?	O.1	5.12.1, I.9	Yes []	No []
MVRP	Is automatic configuration and management of VLAN topology using MVRP supported?	O.1	5.12.1, I.7	Yes []	No []
MRP	Is the Multiple Attribute Registration Protocol (MRP) implemented in support of MRP Applications?	M	10 I.9, I.7, I.8	Yes []	
SPRU	Does the implementation support Source Pruning?	EMRP :O	5.12, 10.10.3, 11.2.1.1	Yes []	No []
MRP1	Does the MRP implementation support operation of the Full Participant?	SPRU:O.2 ¬SPRU:O.3	10 I.8	Yes []	No []
MRP2	Does the MRP implementation support operation of the Full Participant, point-to-point subset?	SPRU:O.2 ¬SPRU:O.3	10 I.8	Yes []	No []
MRP3	Does the MRP implementation support operation of the Applicant-Only Participant?	¬SPRU:O.3	10 I.8	Yes []	No []
MRP4	Does the MRP implementation support operation of the Simple-Applicant Participant, point-to-point subset?	¬SPRU:O.3	10 I.8	Yes []	No []
<u>FQTSSE</u>	<u>Does the implementation support forwarding and queuing for time-sensitive streams?</u>	<u>O</u>	<u>5.18, 34.6</u>	<u>Yes []</u>	<u>No []</u>

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Insert new Table I.9 as follows:

I.9 Forwarding and queuing for time-sensitive streams

Item	Feature	Status	References	Support
FQTSSE1	Support a minimum of two traffic classes, of which one supports the strict priority algorithm and the other is an SR class?	FQTSSE:M	5.18, 8.6.8.1, 34.6	Yes [] N/A []
FQTSSE2	Support the credit-based shaper algorithm as the transmission selection algorithm for frames transmitted for each stream associated with the SR class.	FQTSSE:M	5.18, 8.6.8.2, 34.6	Yes [] N/A []
FQTSSE3	Support the operation of the credit-based shaper algorithm on all Ports as the transmission selection algorithm used for the SR class.	FQTSSE:M	5.18, 8.6.8.2, 34.6	Yes [] N/A []
FQTSSE4	Use the default priority associated with SR class “B” as shown in Table 6-6 as the priority value carried in transmitted SR class “B” data frames.	FQTSSE:M	5.18, Table 6-6, 34.6	Yes [] N/A []
FQTSSE5	Support two or more SR classes (a maximum of seven), and support the operation of the credit-based shaper algorithm on all Ports as the transmission selection algorithm used for those SR classes. The number of SR classes supported shall be stated in the PICS.	FQTSSE:O	5.18, 8.6.8.2, 34.6	Yes [] No [] Number_____
FQTSSE6	Use the default priority associated with SR class “A” as shown in Table 6-6 as the priority value carried in transmitted SR class “A” data frames. If more than two SR classes are supported, the priority value carried in transmitted data frames for the additional SR classes shall be stated in the PICS.	FQTSSE:O	5.18, Table 6-6, 34.6	Yes [] No [] Priority over-ride values_____

Insert the following text and figures as new Annex L:

Annex L

(informative)

Operation of the credit-based shaper algorithm

This annex contains a more detailed analysis of the way the credit-based shaper algorithm (8.6.8.2) operates, and how its operation affects the performance of the network, from the point of view of traffic that uses a credit-based shaped queue, and also from the point of view of other queues associated with the Port.

L.1 Overview of credit-based shaper operation

The credit-based shaper algorithm has a single externally determined parameter, *idleSlope* (see 8.6.8.2), that determines the maximum fraction of the *portTransmitRate* that is available to the queue associated with a traffic class (*bandwidthFraction*), as follows in Equation (L.1):

$$\text{bandwidthFraction} = \text{idleSlope} / \text{portTransmitRate} \quad (\text{L.1})$$

where *portTransmitRate* is the maximum transmit data rate that the port supporting the outbound queue is able to deliver. The detailed derivation of Equation (L.1) can be seen in Equation (L.5) through Equation (L.8).

If a traffic class supported by the credit-based shaper algorithm uses less than the bandwidth allocated to it, then the unused bandwidth can be used by other traffic classes, in accordance with the relative priorities of the traffic classes and the transmission selection algorithms that are associated with them.

NOTE 1—It is a natural consequence of the way transmission selection operates that if a given traffic class does not have a frame available for transmission, then a lower priority traffic class that does have a frame available for transmission is able to transmit. Hence, if a given traffic class that uses the credit-based shaper algorithm has been allocated X% of the available bandwidth on a Port, but only uses (X-Y)%, then the unused portion of its allocation (Y%) is available for other traffic classes to use if they are in a position to do so. Whether this unused bandwidth can be used by a traffic class depends upon the transmission selection algorithm that is associated with it. For example, a traffic class that has been allocated a specific bandwidth, such as one that uses the credit-based shaper algorithm, would be unable to use any bandwidth allocation not used by a higher priority traffic class, as that would result in it exceeding its bandwidth allocation; however, a traffic class that uses the strict priority algorithm, which is not associated with a bandwidth allocation, would be able to use bandwidth allocated to, but not used by, a higher priority traffic class.

The *idleSlope* parameter, in conjunction with the size of the frames that are being transmitted using the queue, and the maximum time delay that a queue can experience before it is able to transmit a queued frame, places an upper bound on the burst size that can be transmitted from queues that use the algorithm.

In order to describe the operation of the algorithm, it is useful to define the following values, in addition to the formal parameters of the algorithm described in 8.6.8.2:

- a) **maxFrameSize.** The maximum sized frame that can be transmitted through the Port for the traffic class concerned. This value can be determined by admission control algorithms, or can be simply a function of the transmission behavior of the traffic source. Either way, the value can be smaller than would be allowed by the normal operation of the underlying MAC service.
- b) **hiCredit.** The maximum value that can be accumulated in the *credit* parameter. This parameter is a function of the operation of the algorithm, and cannot be controlled by management.

- c) ***loCredit***. The minimum value that can be accumulated in the *credit* parameter. The value of *loCredit* is a function of the values of *maxFrameSize*, *portTransmitRate*, and *sendSlope*, as follows in Equation (L.2).

$$loCredit = maxFrameSize \times (sendSlope/portTransmitRate) \quad (L.2)$$

- d) ***maxInterferenceSize***. The maximum size, in bits, of any burst of traffic that can delay the transmission of a frame that is available for transmission for this traffic class. For the numerically highest traffic class, *maxInterferenceSize* is exactly equal to the maximum sized frame that can be transmitted through the Port (the maximum frame size supported by the underlying MAC). For all other traffic classes, the derivation of *maxInterferenceSize* becomes more complex, owing to the possibility that any higher traffic classes that support the credit-based shaper algorithm can delay the transmission of a frame. *maxInterferenceSize* must also account for any media access delay, such as the recovery time in Energy Efficient Ethernet. A detailed analysis can be found in L.3.1.1.

NOTE 2—The definition of *maxInterferenceSize* assumes that any traffic classes that support the shaper algorithm are numerically higher than any traffic classes that support the default strict priority algorithm (8.6.8.1). If this is not true, then the value of *maxInterferenceSize* is infinite, and the operation of the credit-based shaper algorithm cannot provide the bandwidth that has been reserved for it. A detailed analysis of how *maxInterferenceSize* can be determined for a given traffic class is contained in L.3.

The value of *hiCredit* is determined by the worst case interfering traffic, as follows in Equation (L.3).

$$hiCredit = maxInterferenceSize \times (idleSlope/portTransmitRate) \quad (L.3)$$

The value of *hiCredit* can therefore be considered as a “high water mark”; the operation of the algorithm is such that the value of the *credit* parameter will never exceed *hiCredit*.

The choice of value for *idleSlope*, and the calculated value of *hiCredit*, can be used to determine the maximum burst size that can be output from the queue, as follows in Equation (L.4).

$$maxBurstSize = (portTransmitRate \times ((hiCredit - loCredit)/(-sendSlope))) \quad (L.4)$$

NOTE 3—There is an interrelationship between the maximum burst size and the buffer size available to the traffic class. There is no point in accumulating more credit than the traffic class can handle in queued frames, as the limiting factor is the discarding of frames when the queue overflows. This means that the amount of bandwidth that can be reserved on a given Port for a given traffic class depends upon the amount of buffering available to that traffic class. This in turn, combined with the port’s transmission rate, will determine the maximum value of idle slope that can be supported for that traffic class on that Port.

The operation of the credit-based shaper algorithm is illustrated in the examples shown in Figure L-1 through Figure L-3. In Figure L-1, a frame is queued at a time when the *credit* value is zero, and there is no conflicting traffic (there is no higher priority traffic awaiting transmission, and there is no frame being transmitted on the Port). The frame is immediately selected for transmission, and *credit* decreases at the rate of *sendSlope* as the transmission proceeds. Once the frame transmission is complete, *credit* increases back to zero at the rate of *idleSlope*, at which point, a further frame can be selected for transmission.

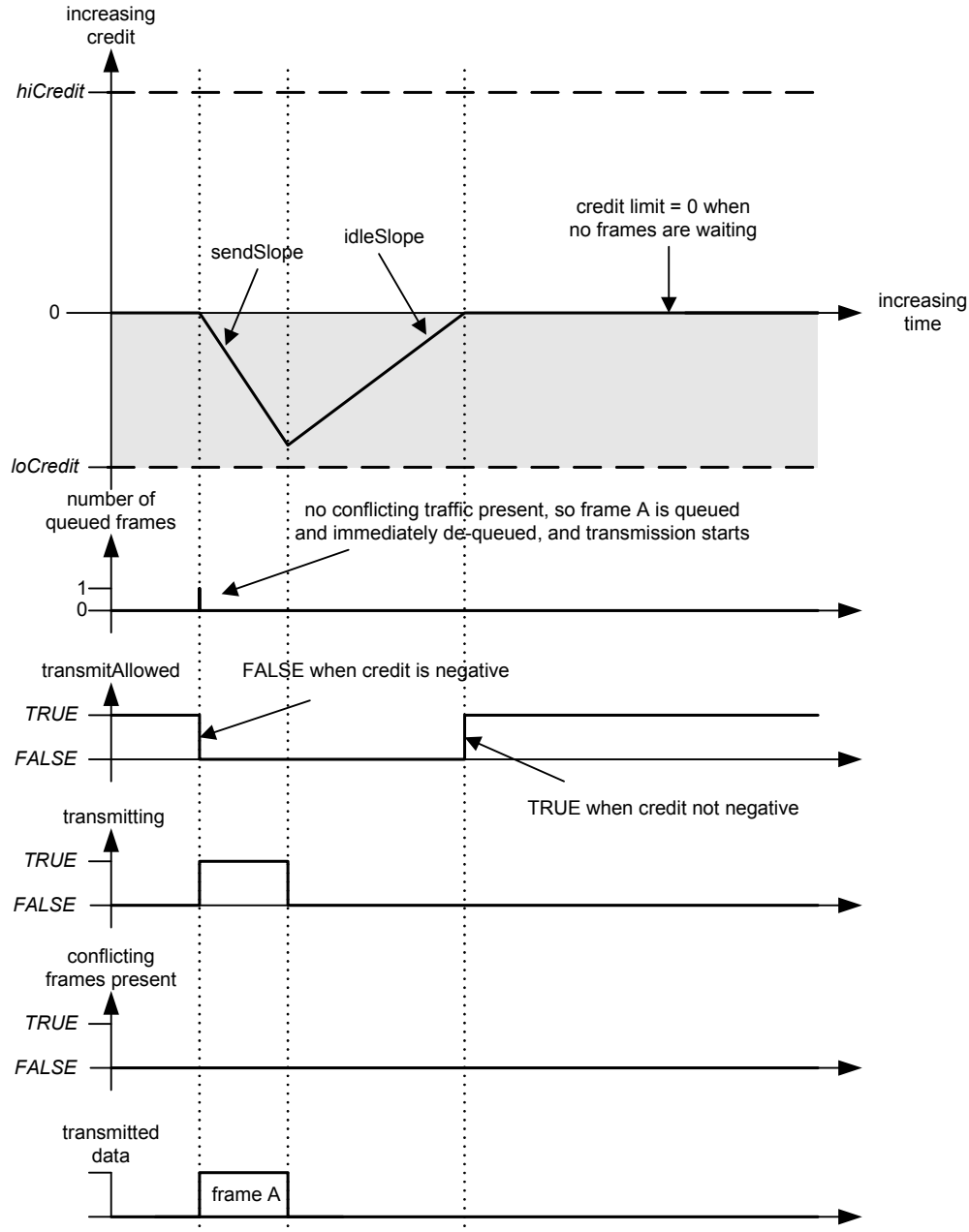


Figure L-1—Credit-based shaper operation—no conflicting traffic

If a continuous stream of frames is made available to the shaper algorithm, i.e., there is always one frame queued awaiting transmission when the *credit* value reaches zero, then the fraction of the *portTransmitRate* that is available to the queue (*bandwidthFraction*) is equal to the fraction of time that frames are being transmitted from the queue. Assuming that *credit* decreases to value *loCredit* at the end of each frame transmission, then the following shows the detailed derivation of Equation (L.1):

$$\text{bandwidthFraction} = (-\text{loCredit}/\text{sendSlope}) / [(-\text{loCredit}/\text{sendSlope}) + (\text{loCredit}/\text{idleSlope})] \quad (\text{L.5})$$

$$= (-1/\text{sendSlope}) / [(1/\text{idleSlope}) - (1/\text{sendSlope})] \quad (\text{L.6})$$

$$= (-idleSlope)/(sendSlope - idleSlope) \quad (L.7)$$

$$= idleSlope/portTransmitRate \quad (L.8)$$

This fraction of the bandwidth is available to the stream even in the presence of conflicting traffic, as illustrated in the following examples. In Figure L-2, a frame is queued at a time when the Port is transmitting conflicting traffic. The value of *credit* increases at the rate of *idleSlope* while the queued frame waits for the Port to become available. Transmission of the conflicting traffic completes before the value of *credit* is limited by *hiCredit*, and transmission of the queued frame starts. The value of *credit* starts to decrease at the rate of *sendSlope*; however, as the frame is not large enough to consume all of the available *credit*, and there are no further frames queued, *credit* is reduced to zero on completion of the transmission.

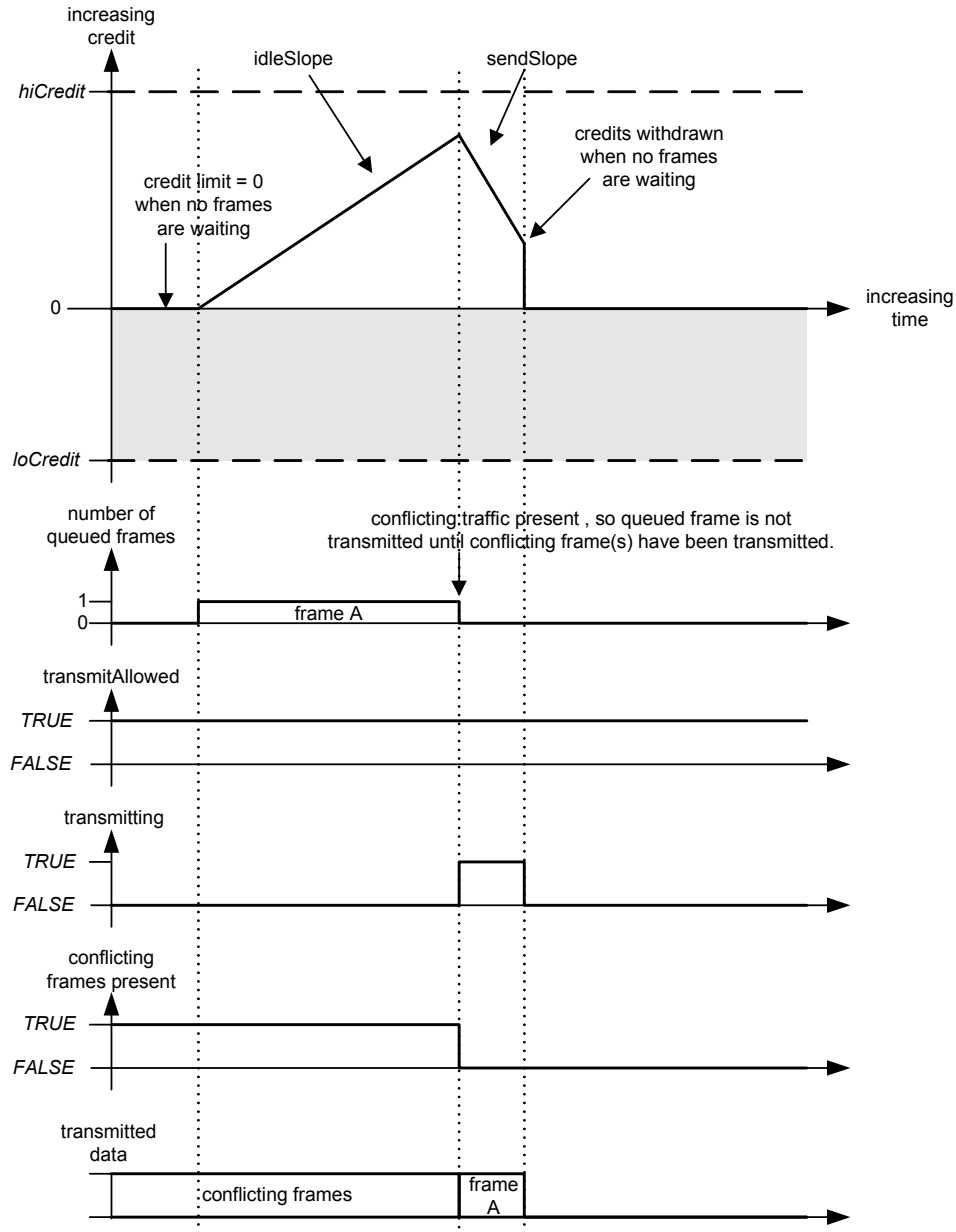


Figure L-2—Credit-based shaper operation—conflicting traffic

In Figure L-3, three frames are queued while the Port is transmitting conflicting traffic, and *credit* accumulates at the rate of *idleSlope*. Once the conflicting traffic has been transmitted, the first and second frames are transmitted back-to-back, because transmitting the first frame leaves $credit \geq 0$. However, as transmitting the second frame causes credit to become negative, transmission of the third frame is delayed until *credit* returns to zero.

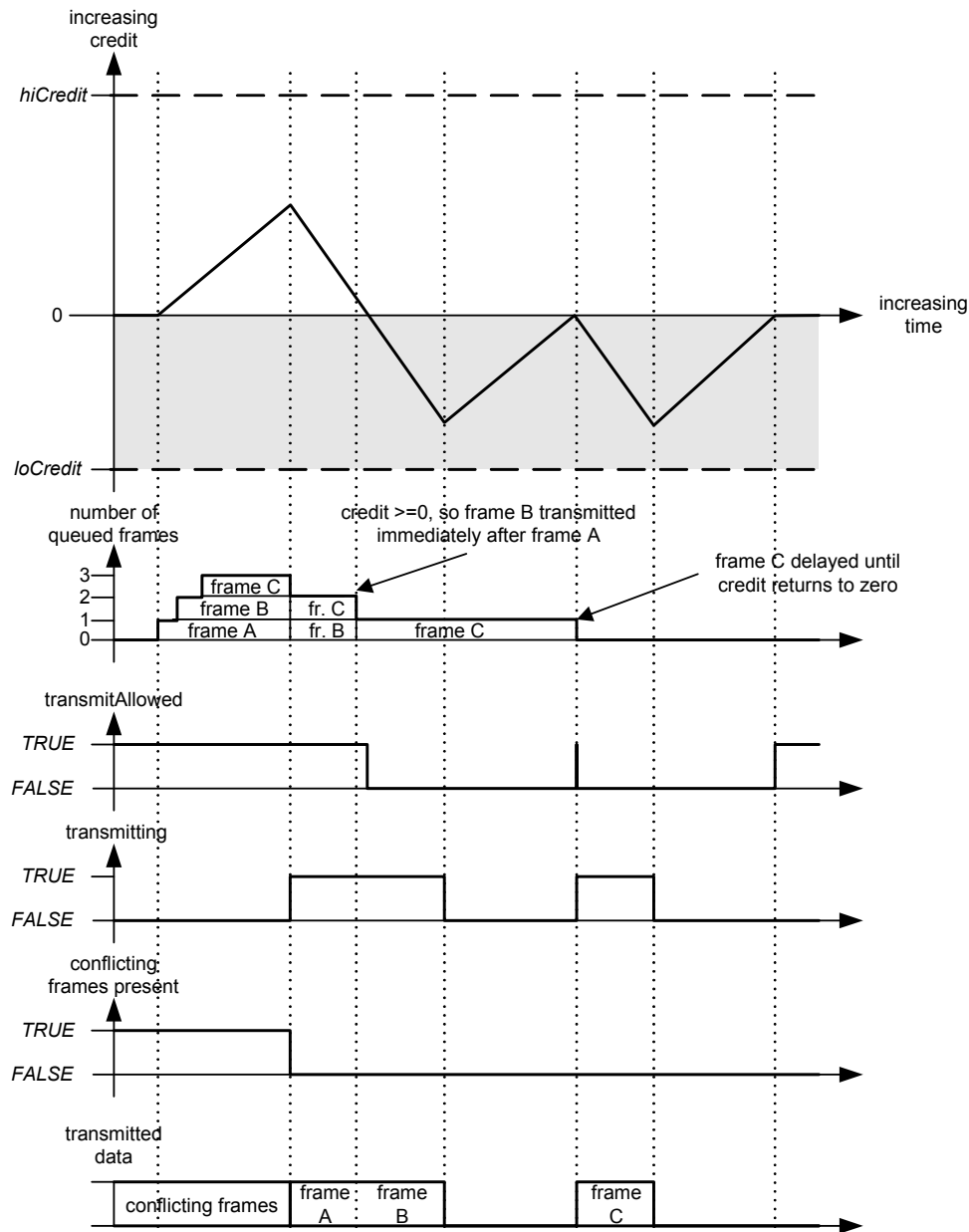


Figure L-3—Credit-based shaper operation—burst traffic

L.2 “Class measurement intervals” in Bridges

The “class measurement interval” defines acceptable Talker behavior with respect to per-stream transmission (see 34.6.1); however, it does not directly impact the operation of the credit-based shaper algorithm in Bridges.

The maximum fraction of the available bandwidth that a queue operating the credit-based shaper algorithm consumes is determined by *idleSlope*, as discussed in L.1; if *idleSlope* is 75% of the link transmission rate, then the algorithm gets a maximum of 75% of the link, averaged over some time period (the class measurement interval) at the Talker. For a Bridge, where transmission becomes bursty due to interfering transmission from other traffic classes, the minimum measurement interval that is meaningful (i.e., the minimum period over which it is still possible to observe that the 75% maximum bandwidth is not exceeded) is the time taken for the traffic class concerned to transmit a maximum sized burst, multiplied by 100/75. In the case of the 125 microsecond class measurement interval for SR class A traffic, on 100 Mbit/s Ethernet, and with 75% reservation, the approximate calculation is (ignoring interframe gaps):

- a) The maximum interference size (*maxInterferenceSize*) is one maximum sized Ethernet frame, which is 2000 octets, or 16000 bits, so *hiCredit* would be 75% of that, or 12000 bits.

NOTE 1—Media access delays, such as those imposed by Energy Efficient Ethernet, could be more significant than the interference created by a maximum sized frame.

NOTE 2—The figure of 2000 octets is a maximum for Ethernet and therefore a worst case for an interfering frame; in a specific implementation, the maximum frame size that is used could be smaller. In implementations where the maximum frame size is known to be less than 2000, the smaller value could be used.

- b) The maximum frame size for this traffic class is 75% of the number of bits that can be transmitted in 125 μ s, so 75% of 12 500 bits in the case of a 100 Mbit/s LAN, or 9375 bits (including up to 500 or so possible “overhead” octets—used for Tag headers, physical layer overhead, etc.). As this is not an integral number of octets, this figure is rounded down to 9368. *loCredit* is 25% of that, or –2342.
- c) The maximum burst size, for frames no greater than 9368 bits long, would occur if *credit* reached 1200, then a series of frames were transmitted that exactly took credit to 0, followed by a final 9368 bit frame that would take credit down to –2342. This maximum burst is therefore calculated as: $(12\ 000 + 2342) \cdot (\text{portTransmitRate}/(-\text{sendSlope}))$ or 57 368 bits.
- d) The minimum measurement interval over which one can expect to be able to measure the bandwidth utilisation for SR class A, and observe that it does not exceed its 75% allocation, is therefore 57368 bit times multiplied by 100/75, which is 76 491 bit times, rounded up to the nearest integer, or 764.91 μ s.

NOTE 3—The maximum sized burst will be preceded by a *maxInterferenceSize* event, and at the end of the burst, *credit* will have been reduced to –2342; it will therefore take $2342/\text{idleSlope}$ seconds, or nearly 3123 bit times, for *credit* to return to zero, and for SR Class A to be allowed to transmit again. In addition, the actual utilization for SR Class A is $57368/(57368+16000+(2342/0.75)) = 75\%$.

For the SR class B traffic, all that really changes is the potential size of any interference, which increases from the SR class A *maxInterferenceSize* to the maximum burst size for SR class A plus the class A *maxInterferenceSize*.

So, the smallest possible measurement interval over which the ratio $\text{idleSlope}/\text{portTransmitRate}$ will be seen to hold good is dependent on the maximum interference size for the SR class, the maximum frame size for the SR class, and the value of *idleSlope*.

L.3 Determining worst-case latency contribution and buffering requirements

From the point of view of the operation of AV Bridges, it is important to be able to assess the contribution that each bridge makes to the worst case latency that stream data frames can be subject to during their transmission along the path from Talker to Listener. Closely related to the latency contribution that a Bridge makes is the buffering requirement that is required in the Bridge in order for the delay not to result in frame discard; if a Bridge can potentially delay a frame of a given SR class for a given period of time, then it must be capable of buffering the worst-case number of frames of that same SR class that could be received for

onward transmission on each Port. The following discussion looks at the factors that contribute to the delay that could be experienced by a frame as it passes through a Bridge, and how the delay can be accurately determined.

NOTE 1—This discussion is also useful to guide designers of Talker stations in determining the delay added by the Talker's network interface.

The worst case latency for a single hop from Bridge to Bridge, measured from arrival of the last bit at Port n of Bridge A to the arrival of the last bit at Port m of Bridge B, can be broken out into the following components:

- a) Input queuing delay. (There are no input queues in the IEEE 802.1 architecture, but if present, the implementation must account for them.)
- b) Interference delay (L.3.1).
- c) Frame transmission delay. (The time taken to transmit one maximum frame at *portTransmitRate*.)
- d) LAN propagation delay. (A variable delay that depends on the length of the LAN connection to the next Bridge; this is measurable using the mechanisms defined in IEEE P802.1AS⁵ [B30].)
- e) Store-and-forward delay. This includes all other elements of forwarding delay that are a consequence of the internal processing of the Bridge, assuming that the input and output queues are empty, such as:
 - 1) The time needed to pass a frame from the input port to the output port, assuming empty queues.
 - 2) The time delay between a frame being available for transmission on a Port and the Port being ready to transmit the frame.

NOTE 2—For example, in the case where the MAC/PHY has entered a power saving mode, there may be a delay incurred in switching the Port back to normal operation.

- 3) The difference, if any, in the delay incurred by a frame that bypasses an empty queue, vs. that incurred by a frame that must be enqueued.
- 4) The time added (subtracted) by the lengthening (shortening) of the frame due to addition (removal) of frame headers such as Q-tags or MACSec-tags.
- 5) The time needed to encrypt a MACSec frame.

In the remainder of this clause, the following abbreviations are used:

M_0	Maximum sized frame for non-SR classes
M_x	Maximum sized frame for SR class x
R_0	<i>portTransmitRate</i>
R_x	<i>idleSlope</i> for SR class x
T_{xy}	Time interval between events x and y
W_x	<i>-sendSlope_x</i>
$variable_{<x}$	The sum of the values of the named <i>variable</i> for all SR classes with a higher priority (and therefore earlier letter in the collating sequence) than x

L.3.1 Interference delay

The interference delay for frame X can be broken out into the following components:

- a) **Queuing delay:** The delay caused by the frame that was selected for transmission an arbitrarily small time before frame X became eligible for transmission selection, plus the delay caused by queued-up frames from all stream frames with higher priority than frame X 's class (i.e., the

⁵IEEE P802.1AS [B30] was a draft standard under development at the time of publication.

“*maxBurstSize*” for SR Classes with higher priority than X—see L.1). This is what is referred to as *maxInterferenceSize* in L.1. Queuing delay is analyzed in detail in L.3.1.1.

- b) **Fan-in delay:** The delay caused by other frames in the same class as frame X that arrive at more-or-less the same time from different input Ports. Fan-in delay is analyzed in detail in L.3.1.2.
- c) **Permanent delay:** The delay caused by frames that reside in a buffer for a long time, relative to the output queuing delay, because of the history of activity in the network. Permanent delay is analyzed in detail in L.3.1.3.

L.3.1.1 Queuing delay

The queuing delay for frame X can be broken out into the following components:

- a) The delay, *maxInterferenceTime*, before a frame X queued for an SR class, and that is eligible for transmission, can actually be transmitted. This delay is experienced when a frame, of any priority, is selected for transmission an arbitrarily small time before frame X became eligible for transmission; it is also experienced in LAN media, such as Ethernet, where there can be a low energy consumption mode that is entered under idle conditions, and where there is a significant delay, *mediumAccessDelay*, before the LAN returns to normal operation and is able to transmit. The value of *maxInterferenceTime* is therefore equal to whichever is the greater of the following:
 - 1) The time taken to transmit the maximum frame size supported by the medium, i.e., $\text{maxFrameSize}_{(m)} / \text{portTransmitRate}$.
 - 2) The *mediumAccessDelay* for the medium.
- b) The delay caused by any queued-up frames associated with the next higher priority SR class than frame X’s SR class (i.e., *maxBurstSize* for the next higher priority SR class), if frame X does not belong to the highest SR class.

For SR class A, only the first of these components applies, as there is no higher SR class.

Suppose that the queue for SR class A is full, and it has accumulated the maximum amount of *credit* (*hiCredit*). Because SR class A frames have priority over all other traffic (even BPDUs), *hiCredit* for SR class A is merely the credit accumulated during the *maxInterferenceTime* required to transmit a lower-priority frame. Until that accumulated credit is exhausted, SR class B (C, D,...) frames cannot be transmitted.

If SR class A were permitted to use 100% of the LAN bandwidth, then the SR class A queue would never catch up, because it would use credit as fast as it was gained, and there would never be a chance for any other traffic class to transmit a frame. If SR class A were permitted to use 99% of the LAN bandwidth, then SR class A can transmit back-to-back frames and the accumulated *credit* would be drained at 1% of the LAN bandwidth until it goes negative; at that point, SR class A cannot transmit further frames until its *credit* returns to a non-negative value, which ensures that at least 1% of the bandwidth is available to other traffic classes.

NOTE—In general, if SR Class A is permitted to use X% of the bandwidth, then only (100-X)% remains available for use by lower priority traffic classes. In practice, as 75% is the default value for the highest percentage of *portTransmitRate* that can be used by all SR classes, there is always at least 25% of *portTransmitRate* available for use by the non-SR traffic classes.

Figure L-4 illustrates this burst transmission behavior, and the effect of *maxInterferenceTime* on the latency of SR class frames. At point *a* on the time line, the queues for SR classes A and B are empty, and a maximum sized frame (M_0 bits long) belonging to a non-SR traffic class starts transmission. An instant later, SR class A and B frames arrive that are all of the maximum size that the classes are currently experiencing (M_A and M_B bits long respectively); however, as there is a frame already being transmitted, they are queued. At point *b*, transmission of the non-SR class frame completes, and the first of a burst of four SR class A frames are transmitted. By point *d*, SR class A’s credit has been taken negative by the transmission of the last of its burst of frames, and therefore no further SR Class A frames can be transmitted until its credit is no

longer negative; this allows SR Class B to transmit a frame. What happens at the end of the class B frame will depend on what frames are queued waiting for transmission, and whether the credit available to either of the SR classes is non-negative.

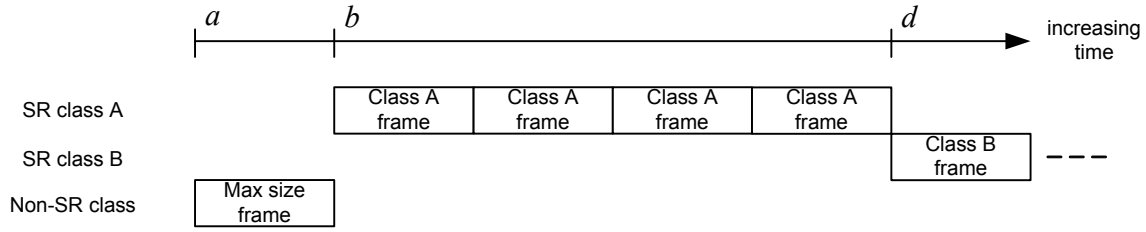


Figure L-4—Interference and latency

The queuing delay experienced by the first SR Class A frame is therefore the time interval between *a* and *b*, T_{ab} . If R_0 is the transmission rate (*portTransmitRate*), then:

$$T_{ab} = M_0 / R_0 = \text{maxInterferenceTime} \quad (\text{L.9})$$

Figure L-5 shows how credit changes for SR Class A during this sequence.

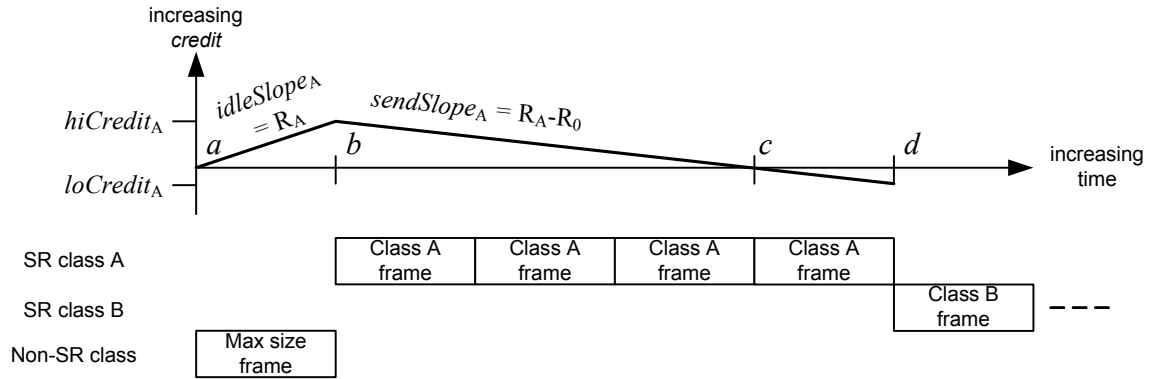


Figure L-5—Burst behavior and credit

R_A is the reserved data rate for SR Class A (which is the *idleSlope* for class A), R_B is the reserved data rate for SR Class B, and so on. The value of $credit_A$ accumulates up to $hiCredit_A$ during the transmission of the maximum sized non-SR frame, so:

$$hiCredit_A = R_A \times M_0 / R_0 = \text{idleSlope}_A \times \text{maxInterferenceTime} \quad (\text{L.10})$$

credit is drained during the transmission of the SR Class A frames, at the rate $sendSlope_A$:

$$sendSlope_A = (R_A - R_0) \quad (\text{L.11})$$

credit reaches zero at point *c* in Figure L-5, which, for the purposes of this discussion, is assumed to happen exactly at the point where transmission of the third class A frame finishes. Since a frame can be transmitted when *credit* = 0, Class A's credits continue to drain to the value:

$$loCredit_A = (R_A - R_0) \times M_A / R_0 \quad (\text{L.12})$$

during the transmission of the fourth class A frame (M_A bits long, transmitted in time M_A / R_0). At point d , transmission of the fourth frame completes; as $credit_A$ is now negative, no more class A frames can be transmitted, and transmission of a class B frame starts.

So, for class A, the maximum burst size is equal to the transmission rate, R_0 , multiplied by the time interval between points b and d in Figure L-5:

$$\text{max Class A burst size} = R_0 \times (- (hiCredit_A - loCredit_A) / sendSlope_A) \quad (L.13)$$

Substituting $hiCredit_A$, $sendSlope_A$, and $loCredit_A$ from Equation (L.10), Equation (L.11), and Equation (L.12) respectively:

$$\text{max Class A burst size} = R_0 \times (- (R_A \times M_0 / R_0 - (R_A - R_0) \times M_A / R_0) / (R_A - R_0)) \quad (L.14)$$

$$= R_0 \times (R_A \times M_0 / R_0) / (R_0 - R_A) + M_A / R_0 \quad (L.15)$$

$$= (R_A \times M_0) / (R_0 - R_A) + M_A \quad (L.16)$$

The queuing delay experienced by SR class B is the time interval between a and d in Figure L-5, T_{ad} :

$$T_{ad} = T_{ab} + T_{bc} + T_{cd} \quad (L.17)$$

$$= M_0 / R_0 + hiCredit_A / sendSlope_A + M_A / R_0 \quad (L.18)$$

$$= M_0 / R_0 + (R_A \times M_0 / R_0) / (R_0 - R_A) + M_A / R_0 \quad (L.19)$$

$$= M_0 / (R_0 - R_A) + M_A / R_0 \quad (L.20)$$

This can be generalized to calculate the queuing delay experienced by any SR class, X , by using the sum of the credits available to all higher priority SR classes. In the following, the subscript “ $<X$ ” is used to indicate the sum of the values for all SR classes with higher priority (and therefore, earlier letters in the collating sequence) than X .

The credit acquisition rate $idleSlope_{<X}$ for SR classes A through $X-1$ is the sum of the data rates for all higher priority classes, so:

$$idleSlope_{<X} = \sum_{K<X} R_K \quad (L.21)$$

This is just the sum of the $idleSlope_K$ values for the higher classes. However, the combined classes accumulate credits only until the single interfering non-SR interfering frame stops transmitting, and then the credits start decreasing linearly. So, the upper limit for the combined credits is not the sum of the individual Class's credits; it is the number of bits divided by the slope, or:

$$hiCredit_{<X} = (\sum_{K<X} R_K) \times M_0 / R_0 \quad (L.22)$$

Similarly, the combined $sendSlope_{<X}$ is:

$$sendSlope_{<X} = - (R_0 - \sum_{K<X} R_K) \quad (L.23)$$

These combined rates hold true until some class's buffer empties and its credits are forced to 0. In the worst-case scenarios we are examining, this does not happen.

Defining

$$W_{<X} = -\text{sendSlope}_{<X} = R_0 - \sum_{K<X} R_K \quad (\text{L.24})$$

we have:

$$\text{sendSlope}_{<X} = -W_{<X} \quad (\text{L.25})$$

and

$$\text{hiCredit}_{<X} = (R_0 - W_{<X}) \times M_0 / R_0 \quad (\text{L.26})$$

For all combined classes, T_{ab} is the same as for SR class A, the time for the non-SR frame to transmit:

$$T_{ab} = M_0 / R_0 \quad (\text{L.27})$$

The combined classes A through X-1 drain from $\text{hiCredit}_{<X}$ to 0 in time $\text{hiCredit}_{<X} / W_{<X}$, so:

$$T_{bc} = ((R_0 - W_{<X}) \times M_0 / R_0) / W_{<X} \quad (\text{L.28})$$

The worst case value of $\text{loCredit}_{<X}$ for the combined $<X$ classes is not the sum $\sum_{K<X} \text{loCredit}_K$ as this would overestimate the worst case. Only one of the $<X$ classes can be transmitting a frame at a given time, so the end of the burst for the $<X$ classes occurs when one of the $<X$ classes, Q say, starts transmitting a frame at a point where all the other $<X$ classes have negative credit, and their credit is therefore rising. If, by the time Q's frame transmission finishes, the other $<X$ classes still have negative credit, then Q's frame is the last frame of the $<X$ burst; but at that point, all the other $<X$ classes must, by definition, have more credit than their respective loCredit values (i.e., for these other classes, $\text{credit} > \text{loCredit}$) because even if they were at their respective loCredit values when Q started transmitting, they have been gaining credit during the transmission of Q's frame.

It is also not possible to decide that $\text{loCredit}_{<X}$ is simply the time needed to transmit one copy of each Class's maximum-length frame; for some classes it could be possible transmit more than a single last frame after the total credits = 0, even if all Classes' credits reach 0 simultaneously. For example, classes A and B reach $\text{credit} = 0$ at the same time; A transmits a frame, then B transmits a frame. By the end of the B frame, A's credit has returned to zero, and so A can transmit a further "last" frame.

Some class must transmit the last frame, and for it to be the worst case, that last frame is a maximum length frame. If it were not, then either:

- c) Extending it to the maximum length still leaves the other classes at negative credit ; or
- d) Extending it to the maximum length leaves one or more other class with 0 or positive credit , in which case they will transmit more frames.

So, for class Q to be the class that transmits the last frame of the $<X$ burst, the following condition must be true for all other $<X$ classes, P:

$$M_q \times R_p > (R_0 - R_p) \times M_p \quad (\text{L.29})$$

Equation (L.29) follows from the fact that the increase in credits of class P during the transmission of M_q , which is equal to $(M_q/R_0) \times R_p$ is greater than the absolute value of the loCredit of P, which is equal to $(R_0 - R_p) \times M_p$. So:

$$(M_q/R_0) \times R_p > (R_0 - R_p) \times M_p / R_0 \quad (\text{L.30})$$

This statement is true because the other classes P must have low enough credit that their credit cannot climb above 0 during the transmission of M_q .

The very lowest that $credit_{<X}$ might reach for three SR classes, Q, P, and S is shown in Equation (L.31).

$$loCredit_{<X} \geq loCredit_Q + loCredit_P + loCredit_S + R_S \times M_q/R_0 + R_P \times M_q/R_0 \quad (L.31)$$

However, even this is pessimistic, as it assumes that more than one class can be at their $loCredit$ at the same time, which is not the case. What this shows, however, is that if R_K is small for all $K < X$ (i.e., $R_K \ll R_0$), then the $R_K \times M_q/R_0$ terms all approach zero, and therefore, $loCredit_{<X}$ approaches the sum $\sum_{K < X} loCredit_K$. So:

$$loCredit_{<X} = \sum_{K < X} loCredit_K \quad (L.32)$$

$$= \sum_{K < X} (R_K - R_0) \times M_K / R_0 \quad (L.33)$$

But in this worst case, $R_K \ll R_0$, so:

$$loCredit_{<X} = \sum_{K < X} (-M_K) \quad (L.34)$$

Putting together these various components, the queuing delay for SR Class X, $qDelay_X$, is as follows:

$$qDelay_X = M_0 / R_0 + (hiCredit_{<X} - loCredit_{<X}) / |sendSlope_{<X}| \quad (L.35)$$

$$= M_0 / R_0 + ((R_0 - W_{<X}) \times M_0 / R_0 + \sum_{K < X} M_K) / W_{<X} \quad (L.36)$$

$$qDelay_X = (M_0 + \sum_{K < X} M_K) / W_{<X} \quad (L.37)$$

For SR class A, this equation reduces to Equation (L.38):

$$qDelay_A = M_0 / W_{<A} \quad (L.38)$$

As there is not a “<A” class, A being the highest priority class, $W_{<A}$ is simply R_0 , so this reduces to Equation (L.39):

$$qDelay_A = M_0 / R_0 \quad (L.39)$$

This agrees with the earlier discussion in L.1.

For SR class B, the equation reduces to Equation (L.40):

$$qDelay_B = (M_0 + M_A) / W_A \quad (L.40)$$

As W_A is $-sendSlope_A$, which is just $R_0 - R_A$, this can be rewritten as shown in Equation (L.41):

$$qDelay_B = (M_0 + M_A) / (R_0 - R_A) \quad (L.41)$$

The maximum burst size for a given class, X, is the number of bits that can be transmitted in “back-to-back” frames by class X. The maximum burst occurs when that class has experienced the maximum queuing delay for that class, $qDelay_X$, and class X is then able to transmit frames, uninterrupted by any higher priority classes, until it reaches $loCredit_X$. The size of this burst, $maxBurst_X$, is equal to the number of bits that can be transmitted at line rate, R_0 , in the time it takes for $credit_X$ to reduce from $hiCredit_X$ to $-loCredit_X$, as follows in Equation (L.42):

$$maxBurst_X = (hiCredit_X - loCredit_X) \times R_0 / W_X \quad (L.42)$$

From the earlier results, the worst case for $loCredit_X$ is just $(-M_X)$, so:

$$= hiCredit_X \times R_0 / W_X + M_X \times R_0 / W_X \quad (L.43)$$

and $hiCredit_X$ is the amount of credit that can be accumulated during $qDelay_X$ seconds, which is $qDelay_X$ multiplied by the idle slope, R_X , so:

$$maxBurst_X = (M_0 + \sum_{K < X} M_K) \times R_X \times R_0 / (W_{<X} \times W_X) + M_X \times R_0 / W_X \quad (L.44)$$

L.3.1.2 Fan-in delay

The delay caused by other frames in the same class as frame X that arrive at more-or-less the same time from different input Ports. Consider the scenario in Figure L-6, where a Bridge has a number of Ports on which it receives frames for a given SR Class and the frames will be queued on a single outbound Port. Imagine that the “upstream” Bridge Ports feeding every reception Port of the Bridge all encounter a maximum interference event for Class X at the same time. The output Port will be starved of data. Then, once the interference events complete, the “upstream” Bridges start to transmit frames and the input Ports all receive a frame at the same moment; all received frames are delivered to the output queue.

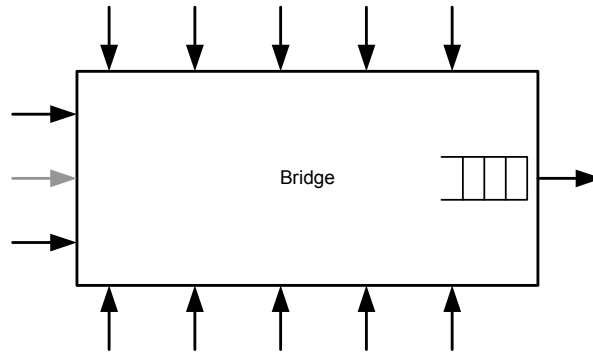


Figure L-6—Fan-in scenario

Assuming that all of the Ports of the Bridge support the same data rate, and the outbound queues of the upstream Bridges are all configured the same as the outbound queue of this Bridge, then the outbound Port would not be able to transmit frames from its queue at the desired, combined data rate of the incoming traffic, as this would require the outbound Port (in this example) to transmit at 20 times its reserved data rate, which would not be permitted by the Stream Reservation Protocol. It follows that whatever the reserved data rate is on the outbound Port, this must be the same as the sum of the reserved data rates being fed into the inbound Ports.

By allocating the majority of the reserved bandwidth to one of the inbound Ports (shown in grey in Figure L-6) and a very small amount of bandwidth to all of the other inbound Ports, the amount of fan-in data that will be received can be maximized; the grey Port will receive $maxBurst_X$ bits of burst data, and all other Ports will receive a single frame of length M_X . Hence, the data that the outbound queue has to handle in this scenario is $maxBurst_X + 12 \times M_X$ bits of fan-in data.

NOTE 1—The reason that this choice maximizes the fan-in data burst size is that regardless of how little bandwidth is allocated to the black Ports, they still get to burst a complete frame.

The following is a procedure (not a formula) for determining the maximum amount of fan-in data for a given SR class and output port:

- Determine B_O , the maximum possible bandwidth for class X on the output port.
- For each possible input port i , determine the maximum possible bandwidth B_i for Class X—it is assumed that this information is obtained by some means, e.g., via LLDP, from the transmitting Bridge that is the source of the frames received by i .

- c) For input port i , calculate $\max\text{Burst}_{X,i}$ using $\max(B_O, B_i)$ for the bandwidth. (In the $\max\text{Burst}$ formula, use $W_X = R_O - \max(B_O, B_i)$.)
- d) Add $\max(\max\text{Burst}_{X,i})$ to the total fan-in data.
- e) Set $B_O = B_O - B_i$ and repeat from step c) until $B_O = 0$.
- f) For each remaining port, add $M_{X,i}$ to the total fan-in data.

The fan-in delay is then equal to the time taken for the total fan-in burst data, computed as shown above, to be output at the line rate of the output port.

NOTE 2—The definition of a measurement interval for stream traffic specifications implies a minimum frame transmission rate (8000 frames per second for class A's 125 us measurement interval). At this frame transmission rate, the frames are smaller and, due to the overhead in small frames, fewer streams are possible (no more than 13 through a 100 Mb port). These constraints should be given due consideration when determining the value to be used for M_X and for determining the maximum number of ports contributing to fan-in delay.

Example calculations of fan-in delay can be found in the paper “Calculating the Delay Added by Qav Stream Queue” [B29].

L.3.1.3 Permanent delay

Permanent delay reflects the fact that, if a path through the network is being used continuously at its full capacity for a particular SR class, interfering traffic can cause “permanent” buffer occupancy at the point of congestion. This is illustrated in Figure L-7 through Figure L-10 and the accompanying text.

In Figure L-7 we have a Talker station, T, that has reserved the highest possible bandwidth, R_X , for an SR class X stream on the path through 3 Bridges to Listener L. For the purposes of the discussion, R_X is the maximum reservable data rate for class X on the outbound Ports of all three Bridges, and Talker T starts transmitting the stream, at the reserved data rate R_X . Assuming that the LANs are short and there is no interfering traffic (i.e., the only traffic being handled is this particular stream), the buffer occupancy in each Bridge will approach zero, as frames can be transmitted onwards as soon as they are received.

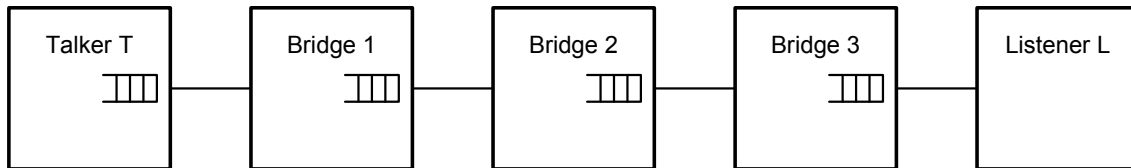
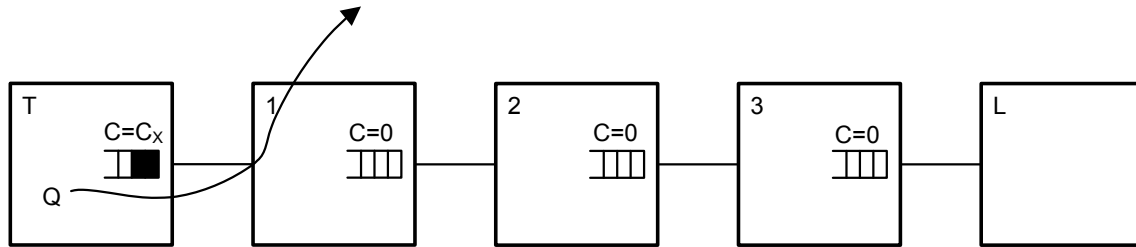
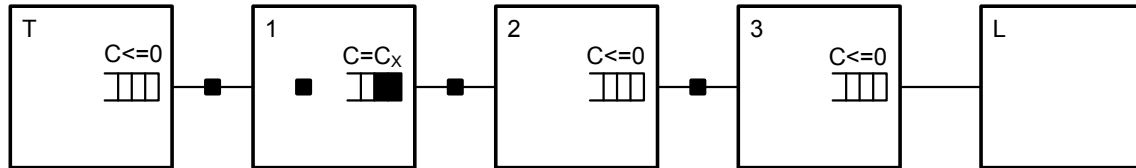


Figure L-7—Permanent delay scenario

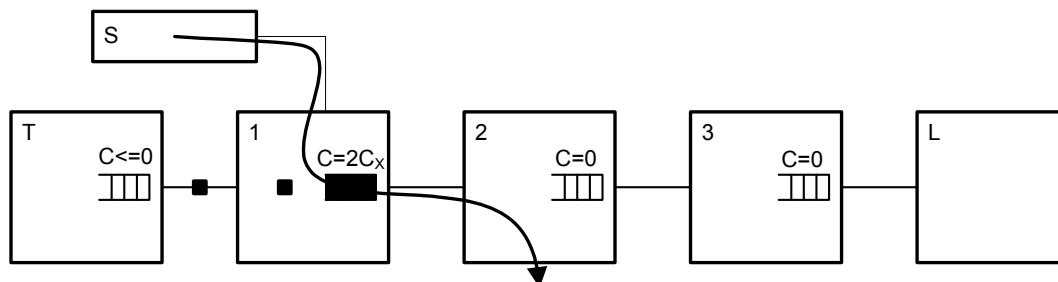
In Figure L-8, a second transmission, Q, generates the maximum possible interference to class X (i.e., class X's next frame for transmission is delayed by $q\text{Delay}_X$). Talker T's class X queue fills, and its *credit* increases to $hiCredit_X$ (shown on the figure as $C=C_X$). Because T is no longer transmitting class X frames, the downstream Bridges are starved of frames to transmit, their buffers are empty, and their credit is zero ($C = 0$).

In Figure L-9, the interfering traffic goes away, and T is able to transmit a maximum sized class X burst of frames at line rate; once the burst has finished, T reverts to transmitting the stream at the reserved rate R_X . As Bridge 1 has no credit, it will have to buffer most of the burst of frames, and as T is sending frames as fast as Bridge 1 can get rid of them, its buffer occupancy will remain at that level unless something else affects that steady state. Bridge 1 ends up with a credit value that hovers around the *hiCredit* level; the credit in the other Bridges will have *credit* of 0 or some value between 0 and *loCredit* as the Bridges receive and transmit frames.

**Figure L-8—Building up buffer occupancy—1****Figure L-9—Building up buffer occupancy—2**

If the interfering traffic, *Q*, starts up again, Talker *T*'s buffer occupancy will build up again; however, as Bridge 1 still has frames buffered, it can continue to transmit, and the downstream Bridges are not starved of frames, and Bridge 1's buffer occupancy and *credit* returns to around 0. When the interference stops, *T* transmits a burst at line rate, as before, which simply restores Bridge 1's queue to the state shown in Figure L-9.

If a new interfering frame source, *S*, starts up that affects class *X* in Bridge 1, then Bridges 2 and 3 could again be starved of traffic, and the buffer occupancy and *credit* in Bridge 1 could increase temporarily to $2C_X$ (see Figure L-10). When *S* stops transmitting, then Bridge 2 can create a line rate burst of traffic that will restore its buffer to the previous steady state (Figure L-9), but now Bridge 2 has a permanent buffer loading as shown in Figure L-10.

**Figure L-10—Building up buffer occupancy—3**

This can of course happen for other Bridges in the path, leading eventually to some level of permanent buffer occupancy in each Bridge. The problem does not get any worse from that point on, because no Bridge ever starves (i.e., no Bridge's *credit* is set to zero because it runs out of frames to send); starvation is a requirement in order for permanent partial occupancy to be created.

It may, therefore, be considered desirable for all Bridges to reserve a little bit more bandwidth than the actual total reserved by the Talkers, so that this “permanent” buffer build-up can drain away. However, this would not make the problem significantly better on the timescales of the output queuing delay, so it does not factor into those calculations, and, of course, this additional capacity would increase the queuing delay at each hop.

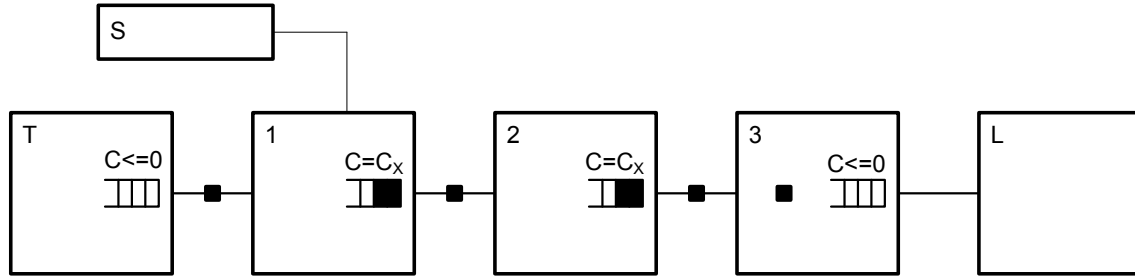


Figure L-11—Building up buffer occupancy—4

NOTE—Many applications have media clocks that are asynchronous to the free-running clock used to determine class measurement intervals. As the two clocks could be operating at the extremes of their permissible variation, it is necessary to reserve more bandwidth than is nominally needed in order to account for the potential difference in clock rate.

Since this kind of event could happen on multiple input ports at the same time, the “permanently buffered” data equals the worst-case fan-in data.

L.3.2 Maximum interference delay and maximum buffer requirement

As discussed in L.3.1, the worst-case interference delay for SR class X on a given Port is the sum of the following three parts:

- a) $qDelay_X$ [see Equation (L.35) in L.3.1.1]
- b) The fan-in delay (see L.3.1.2)
- c) The “permanent” buffer contents contribution (see L.3.1.3), which is equal to the worst-case fan-in delay

The buffering requirements for SR class X consist of the following three components:

- d) A contribution from the queuing delay, equal to $maxBurst_X$
- e) The fan-in burst data size, computed as described in L.3.1.2
- f) The “permanent” buffer contents, which is equal to the fan-in burst data size.

The fan-in burst size and the permanent buffer contents do not count twice; if a fan-in burst happens, then it becomes the permanent buffer contents. If it happens a second time, it can only happen after a pause that has allowed the permanent buffer loading to drain. So the worst case buffering requirement for SR class X is simply the sum of the first two elements, $maxBurst_X$ [Equation (L.42)] and fan-in burst data size, so:

$$maxBuffering_X = maxBurst_X + (fan-in burst data size)_X \quad (L.45)$$

The worst-case total buffering requirement for all of the SR classes on a given Port is the sum of the following two parts:

- g) The worst-case buffering requirement for the lowest-priority SR class on that Port
- h) One max-size frame $M_{i,Z}$ for each class Z of higher priority than Class X for each input port i.

Since the worst case buffer requirement for Class X is when it is taking almost all of the bandwidth from the higher-priority classes, the SR priority levels can share buffering space. The only additional requirement is for the fan-in from higher priorities on other Ports. Thus, the SR classes would do well to share their buffer space on each Port.

L.4 Operation of the credit-based shaper in a coordinated shared network

A Coordinated Shared Network (CSN) is a contention free, QoS capable, time division multiplexed access, network. One of the nodes of the network acts as the Network Coordinator (NC) node granting transmission opportunities to the other nodes of the network. The NC node also acts as the QoS manager of the network.

From the perspective of the specification of the credit-based shaper, a CSN is equivalent to a distributed bridge. Rather than shaping the AV traffic in the egress nodes, which could lack the buffering resources to do so, the shaping could instead be performed by the CSN's Network Coordinator per Class of Service when it schedules transmissions opportunities from the CSN's ingress to the CSN egress nodes. The behavior of CSN's egress to another MAC technology should be consistent with the behavior specified in this standard, regardless of how the shaper is implemented.